Trading against Algorithms: Price Dynamics and Risk Sharing in a Market with Q-learners

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First Draft: August 5, 2025
This Draft: July 14, 2026

Abstract

We study pricing dynamics and risk sharing in a market with rational investors and Q-learning traders. The Q-learners' trading generates a feedback loop in prices: their demand for the risky security depends on their perceived benefit from trading, which in turn, depends on realized returns. We show that this feedback loop generates state-dependent stochastic volatility, predictable returns, and novel price dynamics which depend on the mass and learning rate of the Q-learners. When rational investors have strong risk-sharing motives for trading, we show that Q-learners can (i) earn trading profits and (ii) improve average investor utility, even though they increase the volatility of prices.

JEL Classification: D83, G12, G14, G40

Keywords: Algorithmic trading, Reinforcement learning, Liquidity, Predictability, Human-AI interaction

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1 Introduction

Algorithmic trading is transforming financial markets. An increasing share of trading volume in US stocks is generated by algorithms that are designed to learn and adapt to market conditions in real time.[1] Such strategies are no longer restricted to high-frequency traders or quantitative hedge funds, but are increasingly popular among ETF providers and large asset managers.[2] As a result, regulators are increasingly concerned about algorithmic trading's effect on price volatility, market stability, and shock amplification.[3] For instance, the Financial Policy Committee of the Bank of England warns that[4]

> Advanced AI models could rationally exploit profit-making opportunities in a destabilising way ... these models might identify and exploit weaknesses in the trading strategies of other firms in a way that triggers or amplifies price movements.

We develop a stylized model to answer a fundamental question: How does the presence of algorithmic traders affect equilibrium asset prices, return dynamics, and the welfare of rational investors? Existing simulation-based studies are limited in their ability to address this question because they often restrict attention to economies that are exclusively populated by algorithmic agents.[5] In contrast, our paper provides a first step in understanding human-AI interaction in financial markets. By explicitly modeling how rational investors and algorithms trade against each other, we can characterize the impact of algorithmic trading not only on market outcomes, but also on investor welfare. We show that algorithmic traders can improve average investor welfare even though they earn positive profits and amplify return volatility.

Model. We consider a continuous-time economy populated by a continuum of rational investors with CARA utility, and a mass of algorithmic traders. The market participants

[1]For example, Brogaard, Hendershott, and Riordan (2014) find that algorithms make up 42% of the trading volume in stocks, and Chaboud, Chiquoine, Hjalmarsson, and Vega (2014) find that algorithms make up over 60% of trading volume in some currency markets.

[2]Blackrock, the largest asset manager in the world, introduced active ETFs that are managed by machine learning algorithms in 2018, and more recently announced the introduction of a virtual investment analyst, "Asimov", for use by the firm's fund managers. Traditional money managers, like AQR, are also quickly adopting machine learning-based strategies.

[3]See Regulatory approaches to Artificial Intelligence in finance, OECD Intelligence Papers, No. 24, Sept. 2024, and Financial Stability in Focus: Artificial intelligence in the financial system, Bank of England, Financial Policy Committee, April 2025, for instance.

[4]See Bank of England, Financial Policy Committee, April 2025.

[5]The focus of many of these papers, as we discuss in Section 2, is the microstructure impact of algorithms on market efficiency and liquidity. In contrast, our focus is on understanding the asset pricing and welfare implications of such traders, and as such, we require a model in which algorithms and human investors interact.

trade a risky security in fixed supply which pays dividends following an arithmetic Brownian motion. The rational investors have exposures to a non-tradable endowment risk that is correlated with the dividend: as a result, a fraction have an incentive to be long the risky asset, while the rest have an incentive to be short. The algorithmic traders use Q-learning, which is a foundational reinforcement learning algorithm known for its tractability, economic interpretability, and performance.[6] In the absence of Q-learners, the equilibrium is standard: the price reflects the present value of future dividends, adjusted for a constant risk premium; expected returns are constant, and prices exhibit no predictability or stochastic volatility.

When Q-learners are introduced, the equilibrium changes fundamentally. Rational investors face a nontrivial forecasting problem, because their optimal portfolio choice depends on their expectation of how Q-learners will trade in the future. The Q-learners' future behavior, in turn, depends endogenously on realized returns. With standard Q-learning in discrete time, solving for this equilibrium is not analytically tractable as it involves tracking the evolution of an entire distribution of Q-values. However, by taking the appropriate continuous-time limit, we show that the Q-learners' trading strategies can be summarized by a single state variable, Q_t, which reflects their current estimate of the net benefit to buying a share, and is predictable given the price path.[7]

Feedback loop and equilibrium. When Q_t is positive (negative), Q-learners are more likely to buy (sell) shares of the security. Moreover, a positive return realization increases the estimated net benefit, while a negative return realization decreases it. Because realized returns affect the evolution of Q_t, and consequently, future trading decisions, there is a feedback loop between the security price and the Q-learners' trading. In particular, because rational investors are risk-averse, they require compensation for the induced variation in residual asset supply in the form of a (stochastic) risk premium. This feedback loop between the security price and the Q-learners' trading leads to an amplification of fundamental shocks and is potentially destabilizing. We characterize conditions on the Q-learning algorithm under which there exists an equilibrium, and show that the resulting price function depends nonlinearly on Q_t.

Return dynamics. We show that the equilibrium Q_t process is stochastic and exhibits mean-reversion. As a result, in the presence of Q-learners, security prices exhibit stochastic

[6]See e.g. Wiering and Otterlo (2012), Ch. 1.7.1 or Sutton and Barto (2018), Ch. 6.5.

[7]Specifically, we assume that the Q-learners can only buy or sell up to one share of the risky asset, but (i) use Boltzmann exploration when choosing their demand and (ii) engage in counterfactual learning. As we show in Section 3.3, this allows us to reduce the number of state variables that drive the Q-learners' demand to Q_t. A key technical contribution of the paper is to formally show that the evolution of the Q_t process can be approximated by a stochastic differential equation in continuous time, where the (instantaneous) innovation in the Q_t process is driven by the instantaneous return process. Moreover, as we discuss in Section 4.3, our analysis is robust to relaxing the assumption that the rational investors know the hyperparameters of the Q-learning algorithm.

volatility and predictable returns, even though fundamentals evolve as a Brownian motion. In fact, we show that return volatility is a hump-shaped function of Q_t: it is highest for Q_t near zero (but lower for extreme Q_t), since Q-learner demand is very sensitive to realized returns in this region. In turn, this implies that expected returns are stochastic, predictable, and depend non-monotonically on demand from Q-learners.[8] Finally, while returns exhibit negative autocorrelation at all horizons, the degree of reversals first increases and then decreases with the horizon. This is because shocks to the return process induce moves in Q_t and, consequently, Q-learner demand, that are persistent in the short run, but gradually unwind. We show that the learning rate and the mass of Q-learners differentially affect the strength and persistence of these return dynamics.

Liquidity provision and welfare. A key insight of our analysis is that while algorithmic traders can earn positive profits and amplify return volatility, they can also *raise* average investor welfare. The presence of Q-learners affects investor utility via two channels. First, rational investors are able to exploit the Q-learners' lack of sophistication to extract trading gains: rational investors account for the endogenous demand dynamics induced by Q-learners when making their trades, while Q-learners trade without internalizing equilibrium consequences.

Second, the Q-learners "learn" to provide liquidity to the group of rational investors that demands it more. For instance, suppose the net demand from rational investors is to short the risky security (e.g., if the hedging demand from the short side of the market exceeds the hedging demand from long investors). This excess demand pushes the price lower, and as a result, Q-learners learn that trading alongside long investors is profitable. The impact on welfare is heterogeneous: while short investors benefit from the additional liquidity provision by Q-learners, long investors are worse off due to the added competition. However, there are gains from trade: average investor welfare can be higher despite the fact that Q-learners earn positive profits. This liquidity provision role of Q-learners is consistent with the evidence documented by Hendershott, Jones, and Menkveld (2011), Brogaard et al. (2014) and Boehmer, Fong, and Wu (2021) who show that algorithmic traders improve liquidity, especially for large stocks.

Implications for policy. Two comparative statics results have implications for policy. First, average investor welfare *rises* with the mass of Q-learners, even though it is accompanied by an increase in return volatility. Second, an increase in fundamental volatility can lead to **both** higher Q-learner profits and an increase in average investor utility. Together these results suggest that volatility-based diagnostics of algorithmic-trading harm can be misleading: in our equilibrium, volatility and average welfare move in the *same* direction as

[8]This distinguishes our setting from existing models with noise traders or behavioral investors with extrapolative expectations.

the mass of Q-learners changes, and a blanket restriction on algorithmic participation would reduce risk-sharing capacity rather than improve investor welfare.

Overview. Section 2 discusses related work. Section 3 sets up the model. Section 4 characterizes the equilibrium: the benchmark without Q-learners, a no-hedging benchmark with Q-learners, and the main equilibrium with both. Section 5 derives the return-dynamics predictions. Section 6 presents the welfare and policy analysis. Section 7 concludes. Proofs are in Appendix A and additional analysis is in the Online Appendix.

2 Related literature

Our paper is most closely related to the small but growing literature that focuses on the impact of Q-learners in financial markets.[9] In a representative investor setting, Barberis and Jin (2023) show how a Q-learning algorithm can give rise to commonly observed patterns in investor behavior, including extrapolative demand, experience effects and the inertia in portfolio allocations. Colliard, Foucault, and Lovo (2022) show that a Q-learning market maker in a Glosten and Milgrom-setting adapts to adverse selection due to informed trading, but may charge a markup over the competitive price. They show that noisier environments slow down learning by algorithmic traders thereby leading to less competition and higher spreads. Guarino, Jehiel, and Symons-Hicks (2025) study a similar setting with market makers who use Q-learning. They find that prices vary depending on the setup of the algorithm, and range between competitive and loss-free prices which do not allow trade. Dou, Goldstein, and Ji (2023) introduce Q-learning agents in a multi-trader Kyle-setting and show how such agents can learn to use collusive trading strategies autonomously, by choosing to dampen the sensitivity of their trades to their private information.[10] In related work, Gufler, Sangiorgi, and Tarantino (2025) study how deep reinforcement learning algorithms operate in a financial market that is calibrated to empirically observed levels of return predictability and price impact. Using simulations, they show that algorithmic traders qualitatively match the behavior of rational investors (who have full knowledge of the economy), but fall short quantitatively — the presence of other algorithmic traders slows down learning and leads to lower profits, lower liquidity, and less market efficiency.

[9]There is an earlier literature in finance that uses Q-learning algorithms to numerically solve for equilibria where analytical solutions are infeasible e.g., Goettler, Parlour, and Rajan (2005, 2009).

[10]The latter paper is more broadly related to models of algorithmic collusion in IO settings, including models that focus on Q-learning (e.g., Calvano, Calzolari, Denicoló, and Pastorello (2021), Calvano, Calzolari, Denicolò, and Pastorello (2020), Klein (2021), Banchio and Skrzypacz (2022), Johnson, Rhodes, and Wildenbeest (2023), Xu, Zhang, and Zhao (2024)), richer reinforcement learning algorithms (e.g., Asker, Fershtman, and Pakes (2022) and Cho and Williams (2024)) and large language models (e.g., Fish, Gonczarowski, and Shorrer (2024)).

Relative to this literature, our work differs in two important ways. First, most of the existing analysis is based on simulations of economies with algorithmic agents. In contrast, we provide an explicit characterization of the financial market equilibrium with Q-learning traders, by establishing convergence properties of the Q-learning algorithm in our setting. Our approach is complementary and allows us to transparently describe the key pricing dynamics that result from trading by Q-learners along the entire equilibrium path, and compare our results to the existing theoretical literature.

Second, while existing work has largely focused on the interaction among multiple reinforcement learning agents, our analysis focuses on the interaction between Q-learners and rational investors. As such, our analysis provides a useful benchmark model of human-AI interaction in financial markets. Specifically, our model provides a basic framework to evaluate the impact of algorithmic trading on investor utility and the efficacy of regulatory policy. For instance, we show that the presence of Q-learners in the market can improve average investor utility, even though it leads to higher return volatility and induces nonfundamental dynamics in prices.

Tractably characterizing models with both rational agents and Q-learners is difficult, as rational agents need to predict the Q-learners' behavior, and, therefore, the evolution of Q-values. We render our setting tractable by (i) assuming that Q-learners use counterfactual updating[11] and (ii) taking a continuous-time limit. The latter allows us to characterize the entire equilibrium dynamics via a system of ODEs. That is, we can characterize prices, demands, and trader utilities for any arbitrary history of dividends.

Banchio and Mantegazza (2023) use a similar continuous-time limit to study the emergence of collusive equilibria in a prisoner's dilemma setting. Specifically, they use a "fluid approximation" technique which renders the stochastic Q-value process deterministic in the limit, and which can be interpreted as the mean behavior of the Q-value process. In our continuous-time limit, by contrast, the Q-values follow a stochastic differential equation and are adapted to the filtration generated by prices, since the only relevant shocks are those to dividends. This allows us to capture the inherently stochastic nature of Q-learning while preserving tractability, and renders the rational traders' inference problem tractable.[12]

Dou, Goldstein, Li, and Yang (2025) consider a setting in which better-informed human traders who exhibit k-level thinking compete against an algorithmic trader, and characterize when the algorithm converges to the rational-expectations strategy. The contrast with

[11]Counterfactual updating is a common technique in reinforcement learning, e.g. see Wachter, Mittelstadt, and Russell (2017) and Foerster, Farquhar, Afouras, Nardelli, and Whiteson (2018). In our setting, it means that whenever a Q-learner buys and receives a certain (realized) return, she understands that selling would have yielded minus that return. With this information, the Q-learner can update the Q-values from both buying and selling simultaneously.

[12]See also Cho (2001), which studies asymptotic convergence in an REE setting where agents use least-squares learning.

our setting is sharp on two dimensions. First, instead of assuming a behavioral bias on the human side, we assume rational investors with hedging motives; this lets us define welfare unambiguously[13] and connect the analysis directly to the empirical liquidity-provision literature (Hendershott et al. (2011), Brogaard et al. (2014), Boehmer et al. (2021)). Second, in our equilibrium the Q-learner does *not* converge to a rational-expectations strategy; instead, the presence of Q-learners can endogenously lead to a risk-premium because of the feedback between their trading behavior and prices.

More generally, our paper is related to two other areas in the literature. First, it provides a theoretical complement to the empirical literature that has focused on the impact of algorithmic trading in financial markets (e.g., Brogaard et al. (2014), Chaboud et al. (2014), Weller (2018)). Our model is broadly consistent with the empirical evidence that implies algorithmic traders provide liquidity in markets. Second, it adds to the theoretical literature that focuses on trading settings with non-rational traders. These include models in which rational investors trade against noise traders whose demands are exogenous and uninformed (e.g., Campbell and Kyle (1993), Wang (1993), Wang (1994)) or investors with behavioral biases (e.g., overconfidence, dismissiveness). We show that Q-learners' trading induces a novel source of state-dependent dynamics, because Q-learners systematically and endogenously update their strategies based on past return realizations.[14]

As such, our model is closely related to models of extrapolative investors. The closest comparison is Barberis, Greenwood, Jin, and Shleifer (2015), where sentiment enters demand linearly, yielding constant excess volatility and monotone negative autocorrelation. In contrast, the nonlinear dependence of Q-learner demand on Q_t in our setting delivers stochastic volatility, state-dependent expected returns, and a *non-monotone* term structure of return autocorrelation. As we discuss further in Section 5, these distinctive predictions are consistent with existing empirical evidence.

3 Model

In Section 3.1, we present the key assumptions of the model, Section 3.2 provides a discussion of the important assumptions, and Section 3.3 provides intuition for the process we assume for the Q-learners' trading strategy.

[13]Welfare is difficult to define in settings with heterogeneous beliefs or behavioral biases (e.g., Brunnermeier, Simsek, and Xiong (2014)).

[14]In contrast, traditional models of noise trading assume their security demands are uncorrelated with fundamentals and insensitive to prices, while behavioral investors are usually assumed to maximize subjective utility under biased beliefs or non-standard preferences.

3.1 Setup

Payoffs. There are two securities: a risk free security in perfectly elastic supply with a constant interest rate r, and a risky security which pays a dividend stream:

$$dD_t = \mu dt + \sigma dB_t. \tag{1}$$

The supply of the risky asset is S shares per capita, and the price at date t is given by P_t. Denote the instantaneous return process by

$$dR_t = D_t dt + dP_t - rP_t dt.$$

Rational investors. There is a mass $1-\theta$ of infinitely-lived, rational investors with discount factor δ and CARA utility with risk aversion ϕ. There are two types, indexed by $i \in \{L, S\}$: a fraction $m \in (0,1)$ of mass $1-\theta$ is of type $i = S$, and the remaining fraction $1-m$ is of type $i = L$. Type-i investors are exposed to non-tradable endowment shocks that are correlated with the dividend process. Specifically, an investor of type i chooses consumption C_{it} and risky-asset demand N_{it} to maximize

$$V_i\left(W_0, Q_0\right) = \sup_{\{C_{it}, N_{it}\}_{t\geq 0}} -\mathbb{E}\left[\frac{1}{\phi}\int_0^\infty e^{-\delta t} e^{-\phi C_{it}} dt\right]$$

subject to the wealth dynamics

$$dW_{it} = \left(rW_{it} - C_{it} + N_{it}\left(D_t - rP_t\right)\right) dt + N_{it} dP_t + Y_i\, dB_t, \tag{2}$$

where $Y_S = Y \geq 0$, $Y_L = -Y$, and the parameter Y governs the strength of the rational investors' hedging motives.[15] Intuitively, L investors have an exposure that is negatively correlated with dividend shocks and so have a hedging motive to buy the risky asset, while S investors have positive exposure and so have a hedging motive to short it. Let

$$\bar{Y} \equiv mY - (1-m)\, Y = (2m-1)\, Y$$

denote the rational investors' net hedging exposure. The special case $Y = 0$ shuts down the hedging motive and recovers a model with identical rational investors who trade purely for speculative reasons; this benchmark is nested as a special case of our main equilibrium (Appendix A.1).

Q-learners. There is a mass θ of Q-learners, with aggregate demand N_t^Q for the risky asset,

[15] We do not explicitly model the source of Y_i, but it captures the net risk exposure from the rest of investor i's (non-tradable) portfolio, including shocks to income or liquidity.

where[16]

$$N_t^Q = N^Q(Q_t) = \frac{1-\exp\left(-\frac{1}{\beta}Q_t\right)}{1+\exp\left(-\frac{1}{\beta}Q_t\right)} \in (-1,1), \tag{3}$$

and $Q_t \equiv Q_t^B - Q_t^S$, where Q_t^B and Q_t^S denote the time t Q-values from buying and selling, respectively. As we shall argue below, we can focus on the evolution of a single Q-value, Q_t, which can be expressed as

$$dQ_t = -\alpha Q_t dt + 2\alpha\, dR_t. \tag{4}$$

Here, the parameter $\alpha \in [0,1]$ is the learning rate of the Q-learning algorithm and controls how quickly the algorithm updates Q_t, and the parameter $\beta \geq 0$ captures how responsive the algorithm's demand is to Q_t. Specifically, as β increases the Q-learners' demand N_t^Q responds less to changes in Q_t.

Market clearing and equilibrium. The market clearing condition requires that aggregate demand for the risky asset equals its supply, i.e.,

$$\theta N_t^Q + (1-\theta)\left(mN_t^S + (1-m)\,N_t^L\right) = S.$$

The following defines the notion of equilibrium in our setting.

Definition 1. A Rational Expectations Equilibrium with Q-learning (**Q-REE**) is given by a risk premium $\Pi(Q_t)$, type-i demands for the risky asset $N_{it} = N^i(W_{it}, P_t, D_t, Q_t)$ for $i \in \{L, S\}$, Q-learner demand $N_t^Q = N^Q(Q_t)$, and type-i consumption $C_{it} = C^i(W_{it}, P_t, D_t, Q_t)$, such that

1. Prices: For any (D_t, Q_t) and time $t \geq 0$, the price is given by

$$P_t = P(D_t, Q_t) = \frac{1}{r}\left(D_t + \frac{\mu}{r}\right) + \Pi(Q_t), \tag{5}$$

2. Q-value evolution: Q-values follow Equation (4) given $P(D, Q)$,

3. Utility maximization: for each $i \in \{L, S\}$, $(N^i(\cdot), C^i(\cdot))$ maximizes type-i's expected utility

$$V_i(W_0, Q_0) = \sup_{\{C_{it}, N_{it}\}_{t\geq 0}} -\mathbb{E}_t\left[\frac{1}{\phi}\int_0^\infty e^{-\delta t} e^{-\phi C_{it}} dt\right] \tag{6}$$

subject to the wealth dynamics in Equation (2) and the transversality condition

$$\lim_{\tau\to\infty} \mathbb{E}_t\left[V_i\left(W_i(t+\tau), Q(t+\tau)\right)\right] = 0,$$

[16] We provide a foundation for this functional form in Section 3.3 below. An alternative interpretation is that there is a single large Q-learner who has a misspecified model of price impact. In this case, the demand N_t^Q is simply the Q-learner's demand instead of the aggregate demand.

4. Q-learner demand: $N_t^Q = N^Q(Q_t)$ is given by the expression in Equation (3),

5. Market clearing: for any (D_t, Q_t) and (W_{Lt}, W_{St}), $P(D, Q)$ satisfies

$$\theta N^Q(Q_t) + (1-\theta)\left(mN^S(W_{St}, P_t, D_t, Q_t) + (1-m)\,N^L(W_{Lt}, P_t, D_t, Q_t)\right) = S, \quad (7)$$

6. Rational expectations: Both types of rational investors have correct beliefs about the evolution of $\{D_t, P_t, Q_t\}$.

As we shall establish in Section 4.1, the risk premium Π is constant in the absence of Q-learners, and pinned down by aggregate supply S, risk aversion ϕ, perceived risk σ, and the net hedging exposure $\bar{Y}$ of the rational sector. In the presence of Q-learners, Π depends nonlinearly on the Q-value process Q_t, which itself endogenously depends on $\Pi(Q)$. While Q-learners are not optimizing agents, the rational investors have correct beliefs about the joint distribution of $\{D_t, P_t, Q_t\}$ and optimize accordingly.

3.2 Discussion of assumptions

Our model is stylized for analytical tractability and ease of exposition. Below we discuss the relevance of some of the important simplifying assumptions.

Hyperparameter knowledge. We assume for simplicity that the Q-learning algorithm's hyperparameters α and β as well as the starting Q-value Q_0 are common knowledge. In Section 4.3, we prove that this common knowledge assumption is without loss of generality. That is, if rational investors do not know (α, β, Q_0), they can infer them from observing dividends, prices, and their own demands for an arbitrarily short amount of time in equilibrium. In this sense, equilibrium prices reveal the hyperparameters (almost) instantaneously.

Trading motives of rational investors. The endowment shock $Y_i\, dB_t$ is a reduced-form representation of the non-tradable risk exposures (income, liquidity, or portfolio shocks) that generate hedging demand in standard risk-sharing models (e.g., Wang (1993)). This feature of the model makes the welfare analysis meaningful: without it, rational investors trade only to speculate against Q-learners and would mechanically benefit from their presence. With it, the rational sector has a genuine motive for risk transfer, and Q-learners endogenously earn profits by absorbing part of that risk (see Section 6).

In Appendix E, we consider an alternative setting in which investors trade because they have heterogeneous beliefs about the dividend process. We show that equilibrium behavior is qualitatively similar: in this case, Q-learners can earn profits by trading against the average sentiment of other investors.[17]

[17] We prefer a hedging motive for trade since evaluating welfare in settings with heterogenous beliefs is difficult (e.g., see Brunnermeier et al. (2014)).

Sophistication of algorithmic traders. In practice, investors employing algorithmic trading use proprietary algorithms to guide their trading strategies. Since we do not know specifically which algorithms are used in practice, we assume that algorithmic traders use Q-learning as a simple and transparent stand-in. Q-learning is a foundational reinforcement learning algorithm which is known for its tractability and performance.[18] Q-learning has the additional advantage of being analytically tractable: the updating equation resembles a Bellman equation and we can interpret the Q-value as an estimate for the actual value function.

More generally, however, we expect that many of our results will be qualitatively similar for different reinforcement learning algorithms. Used in a live (or "online") context, such algorithms rely on backward looking data to update a decision rule at each time period. As a result, different algorithms will give rise to a similar feedback loop as described in the introduction: an increase in prices leads the algorithm to buy more, which can then lead to further price increases. We expect that such trading will amplify volatility and generate predictability in equity returns in a similar manner to Q-learning.[19]

Endogenous hyperparameters. Reinforcement learning models such as Q-learning depend on exogenously chosen hyperparameters, i.e., the learning rate α and the temperature parameter β. They are hence subject to the Lucas critique – in reality, firms who delegate trading to an algorithm choose the hyperparameters, so the optimal hyperparameters change depending on the environment.[20] In Appendix C, we endogenize the hyperparameters. There, we assume that a representative trader with CARA utility delegates trading to the Q-learning algorithm and, anticipating the resulting equilibrium, optimally chooses α and β at the outset. The equilibrium takes the same form as in our main model.

3.3 Specification of Q-learning demand

In this subsection, we provide an intuitive argument for why the demand from Q-learners takes the form specified in Equations (3)-(4). We begin with a quick overview of Q-learning in discrete time — Sutton and Barto (2018), Ch. 6.5 provides a more detailed discussion.

Suppose $t = 0, 1, 2, \dots$ and consider the problem of choosing a policy $\{a_t\}_{t\geq 0}$ given a

[18]For textbook treatments of Q-learning, see Wiering and Otterlo (2012), Ch. 1.7.1 or Sutton and Barto (2018), Ch. 6.5. In single-agent dynamic decision problems, Q-learning has been proven to converge to the optimal policy under mild regularity assumptions, see Watkins and Dayan (1992).

[19]For example, a gradient ascent algorithm will increase the probability of buying following positive returns, which in equilibrium further drives up returns. Temporal-Difference (e.g. Sutton and Barto (2018), Ch. 6) or Actor-Critic (e.g. Sutton and Barto (2018), p. 322) algorithms will act similarly.

[20]We are grateful to In-Koo Cho for pointing this out.

stochastically evolving state $\{s_t\}_{t\geq 0}$, where $a_t \in \mathcal{A}$ and $s_t \in \mathcal{S}$ for some finite sets $\mathcal{A}$ and $\mathcal{S}$:

$$V(s_0) = \max_{\{a_t\}_{t\geq 1}} E\left[\sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t)\right].$$

Here, $r(s_t, a_t)$ is the current payoff given (s_t, a_t), γ is the discount factor and $V(s)$ is the value function. The optimal policy and value function can be characterized using the Q-matrix $Q(s,a)$, which is the expected value of choosing action a at state s, assuming optimal play in all future periods, i.e.

$$Q(s,a) = r(s,a) + \gamma E\left[V(s') \mid s, a\right].$$

We then have $V(s) = \max_{a\in\mathcal{A}} Q(s,a)$ and we can exploit this fact to write the Q-matrix recursively as

$$Q(s,a) = r(s,a) + \gamma E\left[\max_{a\in\mathcal{A}} Q(s', a) \mid s, a\right].$$

A Q-learner has no information about the payoff function $r(s,a)$ or the evolution of the state, but instead estimates the Q-matrix recursively: starting with an initial guess for the Q-matrix $\hat{Q}_0$, she updates the Q-matrix via

$$\hat{Q}_{t+1}(s_t, a_t) \leftarrow (1-\alpha)\hat{Q}_t(s_t, a_t) + \alpha\left(r_t + \gamma \max_{a'\in\mathcal{A}} \hat{Q}_t(s_{t+1}, a')\right)$$

whenever action a_t is chosen given state s_t. Here, r_t is the realized payoff at time t, and $\alpha \in [0,1]$ is the learning rate of the Q-learning algorithm, which is set exogenously. It governs how "quickly" the Q-matrix is updated from period to period. In each iteration, the Q-learner takes action a with probability $p(s,a)$.[21]

Now consider a discretization of the model in Section 3.1 in which the risky security pays dividends D_t at time t, where

$$D_{t+1} - D_t = \mu + \sigma\varepsilon_{t+1},$$

and $\varepsilon_{t+1} \sim N(0,1)$ is independent over time.

There is a mass θ of Q-learners indexed by i, each with demand $N_{i,t}^Q \in \{-1, 1\}$ and Q-matrix $Q_{i,t}^a$ for $a \in \{B, S\}$. Here, $Q_{i,t}^B$ is Q-learner i's Q-value from buying a share and $Q_{i,t}^S$ is the Q-value from selling a share.[22] We assume that the Q-learners are identical, i.e.

[21] If a particular value $a \in A$ is not chosen in period t at state s_t, then the Q-matrix is not updated for that value, i.e. $\hat{Q}_{t+1}(s,a) = \hat{Q}_t(s,a)$ whenever $(s,a) \neq (s_t, a_t)$.

[22] We assume that the Q-learners do not treat the dividend as a state. This is natural, since in CARA-Brownian models, the dividend is priced in and returns are independent of realized dividends. Hence, treating the dividend as a state will not improve the Q-learners' payoffs. As we show in Proposition A.1, the return is indeed independent of the dividend D_t in equilibrium.

they have the same hyperparameters α and β and their Q-values are initialized at the same values.

The per period return on the risky security is given by

$$R_{t+1} = P_{t+1} - P_t + D_t - rP_t,$$

and so conditional on buying, each Q-learner updates the Q-value $Q_{i,t}^B$ as follows:

$$Q_{i,t+1}^B = Q_{i,t}^B + \alpha \left(R_{t+1} + \gamma \max_{a \in \{B,S\}} Q_{i,t}^a - Q_{i,t}^B \right), \tag{8}$$

and conditional on selling, she updates $Q_{i,t}^S$ as:

$$Q_{i,t+1}^S = Q_{i,t}^S + \alpha \left(-R_{t+1} + \gamma \max_{a \in \{B,S\}} Q_{i,t}^a - Q_{i,t}^S \right). \tag{9}$$

We make two assumptions on the Q-learning algorithm. First, we assume that the algorithm uses Boltzmann exploration (e.g. Sutton and Barto (2018), p. 37) when choosing demand, i.e.,

$$N_{i,t}^Q = \begin{cases} 1 & \text{with probability } p\left(Q_{i,t}^B, Q_{i,t}^S\right) \\ -1 & \text{with probability } 1 - p\left(Q_{i,t}^B, Q_{i,t}^S\right) \end{cases}$$

where the probability of buying a share $p\left(Q_{i,t}^B, Q_{i,t}^S\right)$ is given by

$$p\left(Q_{i,t}^B, Q_{i,t}^S\right) = \frac{\exp\left(\frac{1}{\beta} Q_{i,t}^B\right)}{\exp\left(\frac{1}{\beta} Q_{i,t}^B\right) + \exp\left(\frac{1}{\beta} Q_{i,t}^S\right)},$$

and where $\beta \geq 0$ is the temperature parameter. Intuitively, a Q-learner is more likely to buy (sell) a share of the risky security when $Q_{i,t}^B > Q_{i,t}^S$ ($Q_{i,t}^B < Q_{i,t}^S$, respectively). The temperature parameter β controls the tradeoff between exploitation and exploration. Specifically, when $\beta \to 0$, the demand function is "greedy" since $p \to \mathbf{1}\left\{Q_{i,t}^B > Q_{i,t}^S\right\}$. On the other hand, when $\beta \to \infty$, the Q-learner randomizes uniformly between buying and selling since $p \to 1/2$.

Second, we assume that Q-learners engage in counterfactual learning. If a Q-learner buys a share today and gets a payoff R_{t+1}, she not only updates $Q_{i,t}^B$ as in Equation (8), but also recognizes that selling would have yielded $-R_{t+1}$ and so simultaneously updates $Q_{i,t}^S$ as in Equation (9). This allows us to gain tractability when solving the model.[23] We can then

[23] For seminal work on the use of counterfactuals in reinforcement learning see Wachter et al. (2017) and Foerster et al. (2018). Our notion of counterfactuals is particularly simple. We only require that the algorithm recognizes that in each period buying and selling yield opposite returns, and that this information is incorporated into Q-values no matter if the algorithm currently buys or sells. In this sense, we assume that the Q-learner is a "price taker."

summarize each Q-learner's behavior via a single Q-value $Q_{i,t} \equiv Q_{i,t}^{B} - Q_{i,t}^{S}$, and characterize the evolution of $Q_{i,t}$ as:

$$\begin{aligned} Q_{i,t+1} &= Q_{i,t} + \alpha \left(R_{t+1} + \gamma \max_{a \in \{B,S\}} Q_{i,t}^{a} - Q_{i,t}^{B} \right) - \alpha \left(-R_{t+1} + \gamma \max_{a \in \{B,S\}} Q_{i,t}^{a} - Q_{i,t}^{S} \right) \\ &= Q_{i,t} \left(1 - \alpha\right) + 2\alpha R_{t+1}. \end{aligned}$$

Since each Q-learner faces the same returns R_{t+1} and updates her Q-matrix the same way, their Q-values are identical, i.e., $Q_{i,t} = Q_t$ for all t, and Q_t follows[24]

$$Q_{t+1} - Q_t = 2\alpha R_{t+1} - \alpha Q_t. \tag{10}$$

We will henceforth refer to Q_t simply as *the Q-value*. With slight abuse of notation, we can express the probability of buying a share as:[25]

$$p\left(Q_t\right) = \frac{\exp\left(\frac{1}{\beta} Q_t\right)}{1 + \exp\left(\frac{1}{\beta} Q_t\right)}.$$

Then, we can express the Q-learners' aggregate demand as

$$N_t^Q = \frac{1}{\theta} \int N_{i,t}^Q di = \mathbb{E}[N_{i,t}^Q | Q_t] = p(Q_t) - (1 - p(Q_t)),$$

where we normalize by θ for convenience. Intuitively, because we have a continuum of Q-learners with the same Q-values, the aggregate demand equals the expected demand for each individual Q-learner.

Using the definition of $p(Q_t)$, we can then write the Q-learners' aggregate demand as

$$N_t^Q = N^Q\left(Q_t\right) = \frac{1 - \exp\left(-\frac{1}{\beta} Q_t\right)}{1 + \exp\left(-\frac{1}{\beta} Q_t\right)} \in \left(-1, 1\right). \tag{11}$$

Figure 1 provides an illustration of the Q-learners' demand as a function of their Q-value, for different levels of β.

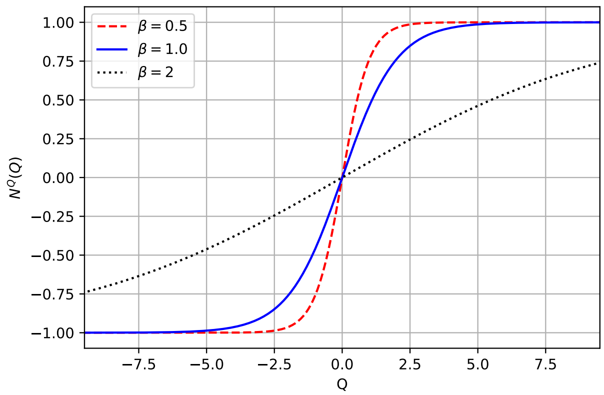
Note that the specification for the Q-learners' demand in Equations (3)-(4) is just the continuous time analogue of the discrete-time specification in Equations (10) and (11). In Appendix B, we establish this convergence formally under the conjectured equilibrium price

[24]Note that Equation (10) is equivalent to the law of motion for $Q_t^B - Q_t^S$ under an expected SARSA algorithm (e.g. Sutton and Barto (2018), Ch. 6.6). Hence, we can alternatively interpret our setting as the algorithmic traders using expected SARSA instead of Q-learning with counterfactual updating.

[25]Since the Q-learning algorithm uses counterfactual updating, the evolution equation (10) is unchanged if we change the algorithm's decision rule, e.g. if we use an ε-greedy rule (e.g. Sutton and Barto (2018), p. 27f) instead of Boltzmann exploration.

Figure 1: Q-learner demand $N^Q(Q_t)$

The figure plots $N^Q(Q)$ as a function of Q for different values of β: $\beta = 0.5$ (dashed), $\beta = 1$ (solid) and $\beta = 2$ (dotted).



function (5).

4 Analysis

In this section we characterize the equilibrium in our model. We proceed in two steps. Section 4.1 shuts down the Q-learners ($\theta = 0$) and characterizes the benchmark equilibrium with the L/S rational sector. Section 4.2 then adds the Q-learners and characterizes the main equilibrium with both Q-learners and a hedging motive. Section 4.3 discusses hyper-parameter uncertainty.

4.1 Benchmark: No Q-learners

The following result characterizes the equilibrium when there are no Q-learners in the economy, i.e., $\theta = 0$.[26]

Proposition 1. *If $\theta = 0$, there exists an equilibrium in which:*

(i) the equilibrium price is

$$P(D_t) = \frac{1}{r}\left(D_t + \frac{\mu}{r}\right) + \Pi_0, \qquad \Pi_0 = -\phi\sigma_{P0}^2 S - \phi\sigma_{P0}\bar{Y},$$

where $\sigma_{P0} = \sigma/r$ and $\bar{Y} = (2m-1)Y$;

[26]In Appendix D, we consider the alternative benchmark in which Q-learners are replaced with a risk-neutral trader who, just as the Q-learners, is restricted to buying or selling at most one share.

(ii) the equilibrium demand for type $i \in \{L, S\}$ is

$$N_{it} = \underbrace{\frac{D_t + \mu/r - rP_t}{r\phi\sigma_{P0}^2}}_{speculative} - \underbrace{\frac{Y_i}{\sigma_{P0}}}_{hedging} = S + \frac{\bar{Y} - Y_i}{\sigma_{P0}};$$

(iii) the value function for type i is

$$V_i(W_t) = -\frac{1}{r\phi}\exp\left(-r\phi W_t - \frac{\delta - r}{r} - \underbrace{\frac{1}{2r}(r\phi)^2(\sigma_{P0}N_i + Y_i)^2}_{gains\ from\ trade}\right),$$

where N_i is the (constant) equilibrium demand from (ii).

The equilibrium price reflects the present value of future dividends, $\frac{1}{r}\left(D_t + \frac{\mu}{r}\right)$, adjusted for two discount components. The term $-\phi\sigma_{P0}^2 S$ is the standard compensation for bearing aggregate supply S. The term $-\phi\sigma_{P0}\bar{Y}$ is the additional discount required by rational investors to absorb the rational sector's net hedging exposure $\bar{Y}$: when $\bar{Y} > 0$ (i.e., $m > 1/2$ and the S side is heavier), the price is lower because the marginal rational investor must be induced to bear additional risk on net. When $\bar{Y} = 0$, the no-hedging case recovers the standard CARA benchmark.

The optimal demand consists of two components. The "speculative" term is the standard mean-variance demand and is common across the two types. The "hedging" term $-Y_i/\sigma_P(Q)$ implies that L investors ($Y_L = -Y$) buy more of the asset and S investors ($Y_S = Y$) sell more to lay off their non-tradable exposure. The value function reflects a gains from trade component, which is quadratic in the investor's net risk exposure $\sigma_{P0}N_i + Y_i$: each type gains whenever the price departs from the present value of discounted cash flows, irrespective of whether it holds a net long or short position.

Finally, note that the price volatility is constant and expected returns do not exhibit predictability since

$$\sigma_{P0} = \frac{\sigma}{r}, \quad \text{and} \quad \mathbb{E}[dR] = r\phi\sigma_{P0}^2 S + r\phi\sigma_{P0}\bar{Y}.$$

As we shall see in the next section, in the presence of Q-learners, price volatility is state dependent. This implies that the risk premium is state-dependent and expected returns are predictable.

4.2 Equilibrium with Q-learners

We now characterize the equilibrium in the general case, where rational investors have heterogeneous hedging needs. We focus (without loss of generality) on $m > 1/2$, so that the rational sector's net hedging exposure $\bar{Y} = (2m-1)\,Y > 0$ and its aggregate hedging demand is to short the asset. The special case $Y = 0$, in which both rational types are identical ($\bar{Y} = 0$) and trade purely for speculative reasons, is nested as $Y \to 0$; we relegate its (simpler) characterization to Appendix A.1.

The following result characterizes the equilibrium.

Proposition 2. *There exists a Q-REE in which the equilibrium price is given by*

$$P_t = \frac{1}{r}\left(D_t + \frac{\mu}{r}\right) + \Pi\left(Q_t\right).$$

The gains from trade for types $i = L, S$ satisfy the ODEs

$$\begin{aligned} rG_i\left(Q\right) \quad = \quad & \tfrac{1}{2}\left(\phi r\right)^2 N_i^2\left(Q\right)\sigma_P^2\left(Q\right) - \tfrac{1}{2}\left(\phi r Y\right)^2 - G_i'\left(Q\right)\phi r \sigma_Q\left(Q\right) Y_i \\ & + G_i'\left(Q\right)\mu_Q\left(Q\right) + \left(G_i''\left(Q\right) - G_i'\left(Q\right)^2\right)\tfrac{1}{2}\sigma_Q^2\left(Q\right) \end{aligned}$$

with boundary conditions $G_i'\left(-\infty\right) = G_i'\left(\infty\right) = 0$. The risk premium satisfies the ODE

$$\begin{aligned} r\Pi\left(Q\right) \quad = \quad & \Pi'\left(Q\right)\mu_Q\left(Q\right) + \tfrac{1}{2}\Pi''\left(Q\right)\sigma_Q^2\left(Q\right) \\ & -\tfrac{\phi r \sigma_P^2\left(Q\right)}{1-\theta}\left(S - \theta N^Q\left(Q\right)\right) - \bar{G}'\left(Q\right)\sigma_P\left(Q\right)\sigma_Q\left(Q\right) - \phi r \bar{Y}\sigma_P\left(Q\right) \end{aligned}$$

on $\mathbb{R}$ with boundary conditions $\Pi'\left(-\infty\right) = \Pi'\left(\infty\right) = 0$, where $\bar{G}'\left(Q\right) = mG_S'\left(Q\right) + \left(1-m\right)G_L'\left(Q\right)$. The evolution of Q is driven by

$$\mu_Q\left(Q\right) = \frac{2\alpha}{1 - 2\alpha\Pi'\left(Q\right)}\left(\frac{1}{2}\sigma_Q^2\left(Q\right)\Pi''\left(Q\right) - r\Pi\left(Q\right) - \frac{1}{2}Q\right), \quad \sigma_Q\left(Q\right) = \frac{\sigma}{r}\frac{2\alpha}{1 - 2\alpha\Pi'\left(Q\right)}.$$

We sketch the heuristic argument behind Proposition 2; the formal proof is in Appendix A. Suppose the price can be characterized as in Equation (5) and let

$$dP_t \equiv \mu_P\left(Q_t\right)dt + \sigma_P\left(Q_t\right)dB_t. \tag{14}$$

Since the flow reward for the Q-learners is $dR_t = \left(D_t - rP_t\right)dt + dP_t$, this implies we can express the evolution of Q as

$$dQ_t \equiv \mu_Q\left(Q_t\right)dt + \sigma_Q\left(Q_t\right)dB_t, \tag{15}$$

where the price conjecture above implies:

$$\mu_Q\left(Q_t\right) = 2\alpha\left(\mu_P\left(Q_t\right) - \frac{\mu}{r} - r\Pi\left(Q_t\right) - \frac{1}{2}Q_t\right), \text{ and } \sigma_Q\left(Q_t\right) = 2\alpha\sigma_P\left(Q_t\right). \tag{16}$$

Using Ito's lemma, we can express

$$\mu_P\left(Q_t\right) = \frac{\mu}{r} + \Pi'\left(Q_t\right)\mu_Q(Q_t) + \frac{1}{2}\Pi''\left(Q_t\right)\sigma_Q^2\left(Q_t\right), \text{ and } \sigma_P\left(Q_t\right) = \frac{\sigma}{r} + \Pi'\left(Q_t\right)\sigma_Q\left(Q_t\right). \tag{17}$$

Next, conjecture that the value function for a type-i rational investor is of the form:

$$V^i(W_t, Q_t) = -\frac{1}{r\phi}\exp\left(-\phi r W_t - \frac{\delta - r}{r} - G_i(Q_t)\right).$$

We refer to $G_i(Q_t)$ as the gains from trade for type i, because it captures the incremental value that accrues to a type-i investor from predicting the demand of the Q-learners and from sharing endowment risk with the other rational type.

The HJB equation for a type-i trader (Equation (39) in the Appendix) accounts for the investor's endowment exposure through the combined risk $N_i\sigma_P + Y_i$. Its first-order conditions imply that optimal consumption is

$$C_{it} = rW_{it} + \frac{\delta - r}{r\phi} + \frac{1}{\phi}G_i(Q_t).$$

The additional value from trading against the Q-learners (reflected in G_i) raises consumption rather than asset demand, since the traders have CARA utility. The optimal risky-asset demand for a type-i investor consists of three components:

$$N_i(Q_t) = \underbrace{\frac{D_t + \mu_P(Q_t) - rP_t}{\phi r\sigma_P^2(Q_t)}}_{\text{speculative}} - \underbrace{\frac{Y_i}{\sigma_P(Q_t)}}_{\text{hedging}} - \underbrace{2\alpha\frac{G_i'(Q_t)}{\phi r}}_{\text{Q-anticipation}}. \tag{18}$$

The speculative and hedging components mirror those in the benchmark without Q-learners, but reflect the change that the drift and volatility of the price process depend stochastically on Q_t. The "Q-anticipation" term reflects how each type adjusts demand to exploit the dynamics of Q-learner demand. Imposing market clearing then yields the equilibrium price

$$P_t = \frac{1}{r}\left(D_t + \mu_P(Q_t) - \sigma_P^2(Q_t)\left(\frac{r\phi}{1-\theta}\left(S - \theta N^Q(Q_t)\right) + 2\alpha\bar{G}'(Q_t)\right) - \phi r\bar{Y}\sigma_P(Q_t)\right),$$

which ties the risk premium to the rational sector's average exposure $\bar{Y}$ via the last term: when $\bar{Y} > 0$, the price is low relative to $Y = 0$ to compensate the net providers of risk sharing.

The key challenge is to establish existence of the ODE system (12) and (13). One cannot rely on standard Lipschitz conditions, since the volatility $\sigma_Q(Q)$ may potentially explode as $\Pi'(Q) \to 1/2\alpha$. Instead, we use sub- and supersolutions to construct bounded solutions on arbitrary finite domains, and then extend these solutions to infinity via the Arzela-Ascoli theorem.

4.2.1 The impact of Q-learning on the risk premium and gains from trade

To build intuition for the equilibrium, first consider the limit $\alpha \to 0$, in which the Q-learners' demand is frozen at $N^Q(Q_0)$. Price volatility is then constant, $\sigma_P = \sigma/r$, and the equilibrium reduces to the no-Q-learner benchmark of Proposition 1, with the asset supply adjusted for the Q-learners' (constant) demand. The risk premium is

$$\Pi(Q) = -\frac{\phi\sigma_P^2}{1-\theta}\left(S - \theta N^Q(Q)\right) - \phi\sigma_P\bar{Y}, \tag{19}$$

and the gains from trade for type i are

$$G_i(Q) = \frac{1}{2r}(r\phi)^2\left(\sigma_P N_i(Q) + Y_i\right)^2,$$

where $N_i(Q)$ is the type-i demand from Equation (18). The first term in (19) is the standard compensation for the net supply the rational sector must hold, adjusted for the Q-learners' demand; the second is the additional discount for absorbing the net hedging exposure $\bar{Y}$. As a result, the risk premium $\Pi(Q)$ is increasing in the net Q-value from buying, Q. Taking $Q \to \pm\infty$ (so that $N^Q \to \pm 1$) recovers the boundary values of Π and G_i in Proposition 2. As in the benchmark, G_i is U-shaped in the type's net risk exposure $\sigma_P N_i + Y_I$.

When $\alpha > 0$, the Q-learners update their Q-value Q_t and, consequently, their demand $N^Q(Q_t)$ for the risky asset, based on realized returns. This induces a feedback channel to prices, whereby larger return realizations lead Q-learners to increase their estimate of the net benefit from buying (versus selling), Q_t, which leads to higher demand $N^Q(Q_t)$, which then leads to higher expected returns.[27] Moreover, this feedback effect cannot be too large — a necessary condition for the existence of an equilibrium is that the rate of learning by the Q-learners is bounded above i.e., $\alpha < \frac{1}{2\Pi'(Q_t)}$. To see why, note that we can express price volatility as

$$\sigma_P(Q_t) = \frac{\sigma}{r} + \frac{\sigma}{r}\frac{2\alpha\Pi'(Q_t)}{1-2\alpha\Pi'(Q_t)} = \frac{\sigma}{r}\frac{1}{1-2\alpha\Pi'(Q_t)}.$$

This implies that, as $\alpha \to \frac{1}{2\Pi'(Q_t)}$, $\sigma_P(Q_t) \to \infty$, and consequently, the rational traders' demand is insensitive to price (see equation (18)). As a result, there is no finite price P_t which clears the market.[28] We establish that $\alpha < \frac{1}{2\Pi'(Q_t)}$ as part of the proof of Proposition 2.

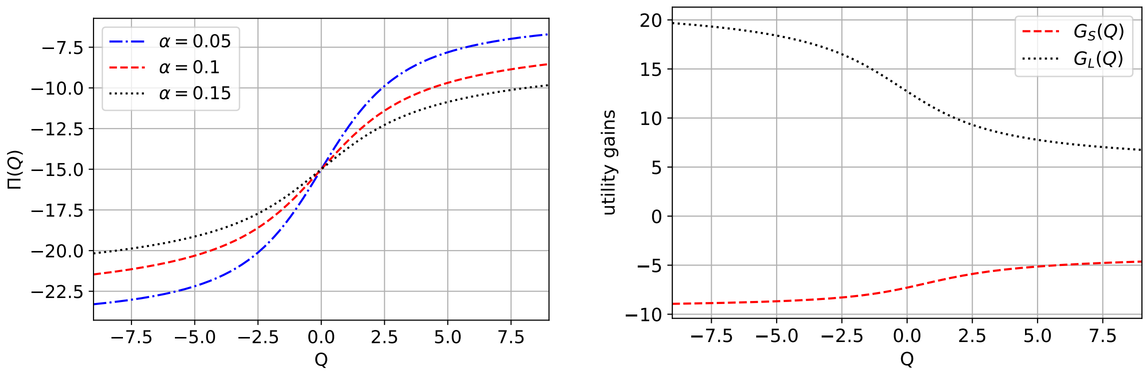
Figure 2 provides a numerical illustration. Panel (a) shows that the risk premium $\Pi(Q)$ is increasing in Q and steeper for larger α, reflecting the stronger feedback channel. Panel (b) shows the two types' gains from trade: because $\bar{Y} > 0$, the S investors bear more risk

[27] This feedback channel is analogous to those in models of positive-feedback or extrapolative investors (e.g., De Long, Shleifer, Summers, and Waldmann (1990), Barberis et al. (2015)).

[28] Moreover, when $\alpha > \frac{1}{2\Pi'(Q_t)}$, price volatility is negative.

Figure 2: Risk premium $\Pi(Q)$ and gains from trade

The figure plots the risk premium $\Pi(Q_t)$ (for different values of the learning rate α) and the gains from trade $G_S(Q_t), G_L(Q_t)$ for the two rational types (at baseline $\alpha = 0.10$). Other parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $S = 0$, $Y = 2$, and $m = 0.75$.



(a) Risk premium $\Pi(Q)$ (b) Gains from trade $G_S(Q), G_L(Q)$

on net and their gains G_S lie below the L investors' gains G_L.

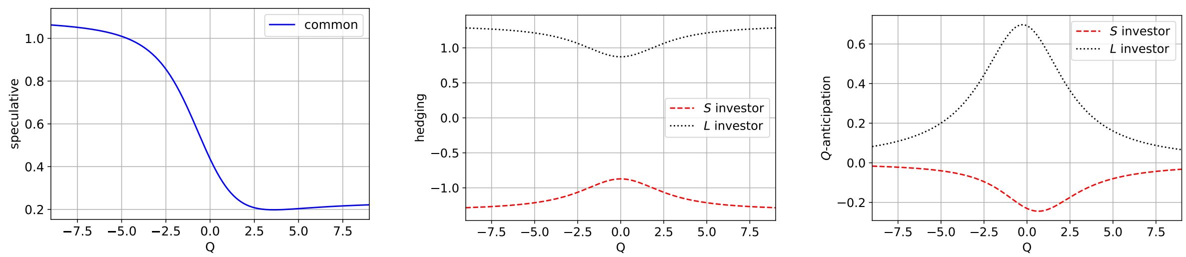
Importantly, the optimal demand of rational investors depends *nonlinearly* on the state-variable Q_t. Figure 3 illustrates the three components of demand in expression (18). The speculative component (panel (a)) is common across types and inherits its nonlinearity from the drift μ_P and volatility σ_P of the price process, which are nonlinear in Q_t. The hedging component (panel (b)) is equal and opposite for the two types, $\pm Y/\sigma_P(Q)$, and varies with Q_t only through price volatility. The Q-anticipation component (panel (c)) is also nonlinear, because rational investors anticipate the dynamics of the Q-learners' demand and adjust accordingly; it differs across types since each type's gains from trade G_i differ. Importantly, this nonlinearity distinguishes our model from most standard models with CARA investors (e.g., Wang (1993), Barberis et al. (2015)), where demand is linear in the relevant state variables. As a result, our model gives rise to stochastic volatility in returns, while existing models exhibit constant or deterministic volatility (as we shall discuss below).

4.2.2 The equilibrium evolution of Q_t

The drift $\mu_Q(Q)$ and volatility $\sigma_Q(Q)$ take the same functional form as in the benchmark, because they depend on Y and m only indirectly, through their effect on Π. Figure 4 illustrates how these vary with the rate of learning, α, and the mass of Q-learners, θ. As panels (a) and (b) suggest, the drift $\mu_Q(Q)$ is decreasing in Q, so that Q_t is mean-reverting. To see why this is intuitive, suppose Q_t is positive and large. This implies Q-learners are more likely to buy shares of the risky security, which pushes up its price. But this makes it

Figure 3: Rational investor demand components versus Q_t

The figure plots the three components of type-i rational demand (as in expression (18)) for the S and L investors. Parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\alpha = 0.10$, $\theta = 0.30$, $S = 0$, $Y = 2$, and $m = 0.75$.



(a) Speculative (b) Hedging (c) Q-anticipation

less profitable to buy the security going forward, which decreases the net value from buying shares Q_t. Moreover, consistent with this intuition, the response is larger when either the rate of learning α is higher or the mass of Q-learners θ is larger. A higher α implies that Q-learners update Q_t more quickly in response to price changes induced by their past behavior, while a larger θ implies that changes in Q_t have a larger impact on prices, which in turn, leads to more mean-reversion.

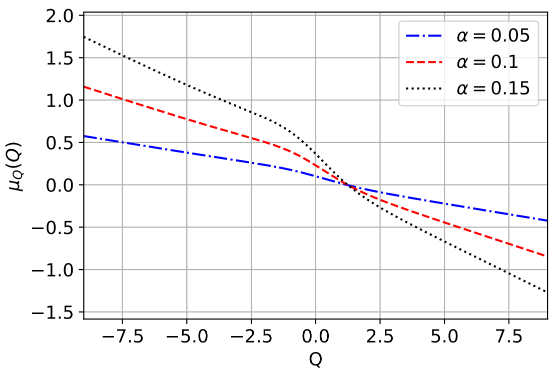
Because $\bar{Y} > 0$, the rest point of Q_t (where $\mu_Q = 0$) is positive: Q-learners tilt long on average, as they learn to absorb part of the rational sector's net hedging demand. A positive $\bar{Y}$ pushes the price lower, which makes buying systematically profitable; the Q-learning algorithm learns this from realized returns and updates Q_t upward, so that the steady state of Q_t is positive and the equilibrium average position of Q-learners tilts long.[29] In effect, Q-learners learn to compete with the L investors in providing liquidity to the S investors, and earn profits for doing so. As we shall see in Section 6, this liquidity provision role of Q-learners can raise average rational investor welfare even while it raises return volatility.

Panels (c) and (d) illustrate that $\sigma_Q(Q)$ is hump-shaped in Q_t and highest when $Q_t = 0$. Intuitively, when Q_t is zero, the Q-learners are indifferent between buying and selling, and so their beliefs are maximally sensitive to realized returns. In contrast, when Q_t is extremely positive (negative), Q-learners perceive the net benefit from buying (selling, respectively) to be extremely high and so Q_t is not very sensitive to realized returns, and consequently, $\sigma_Q(Q)$ is very low. Moreover, all else equal, $\sigma_Q(Q)$ increases as the rate of learning α or the mass of Q-learners θ increases, since either of these changes make Q_t more sensitive to realized returns. The feedback effect induced by Q-learners thus implies that their demand for the risky security evolves in a stochastic but persistent manner. As we shall describe in Section 5, this has novel implications for return dynamics in our setting.

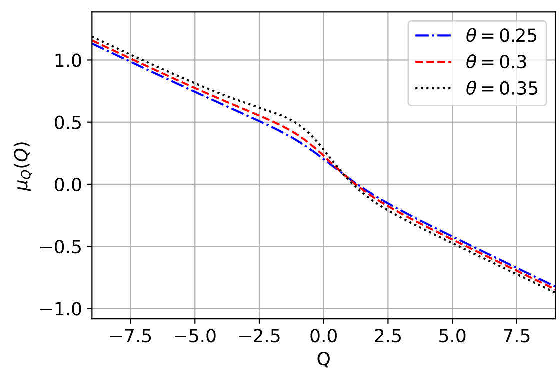
[29] If $\bar{Y} < 0$, then the rest point of Q_t would be negative.

Figure 4: Drift and volatility of the Q_t process

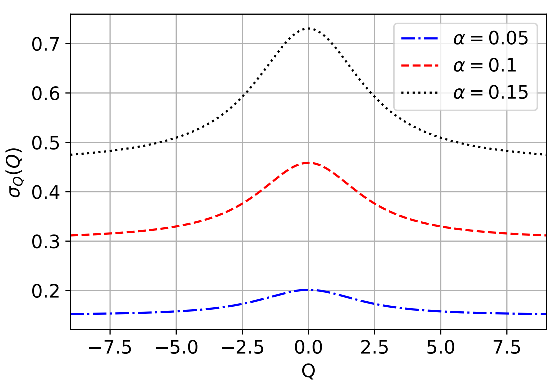
The figure plots the drift $\mu_Q(Q)$ and volatility $\sigma_Q(Q)$ for the Q_t process, for different values of learning rate α and mass of Q-learners, θ. Other parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $Y = 2$, and $m = 0.75$.



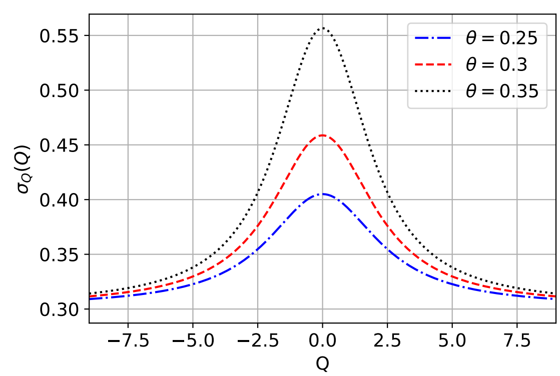
(a) Drift $\mu_Q(Q)$ versus α



(b) Drift $\mu_Q(Q)$ versus θ



(c) Volatility $\sigma_Q(Q)$ versus α



(d) Volatility $\sigma_Q(Q)$ versus θ

4.3 Hyperparameter uncertainty

The baseline model assumes that rational investors know the Q-learning hyperparameters (α, β, Q_0). The following result shows that this is without loss of generality: by observing prices, demands, and realized returns over an arbitrarily short horizon, rational traders can back out the hyperparameters from the implied excess volatility.[30]

Proposition 3. *If traders do not know* (α, β, Q_0), *then the Q-REE of Propositions 2 remains an equilibrium.*

A separate question is whether the algorithmic trader who delegates to the Q-learner would optimally choose those hyperparameters. We address this Lucas-critique concern in

[30] We maintain throughout that traders know the algorithm uses Q-learning with a Boltzmann demand rule and counterfactual updating; they need not know the parameter values.

Online Appendix C, where an algorithmic trader with CARA utility chooses (α, β) optimally and the equilibrium takes the same form as in Proposition 2.

5 Return dynamics

The Q-learners' demand depends nonlinearly on their net benefit Q_t from buying the security. As the previous section demonstrates, this implies that the presence of Q-learners induces stochastic volatility and predictability in expected returns. We explore how these depend on the parameters of Q-learning in this section. Recall that **dollar** returns evolve according to

$$dR_t = (D_t - rP_t)\, dt + dP_t,$$

where P_t satisfies Equation (5).[31] This implies that returns can be characterized as follows.

Corollary 1. *In equilibrium, returns evolve according to:*

$$dR_t = \sigma_P^2 (Q_t) \left(\frac{r\phi}{1-\theta} \left(S - \theta N^Q (Q_t) \right) + 2\alpha G' (Q_t) \right) dt + \sigma_P (Q_t)\, dB_t.$$

where

$$\sigma_P (Q_t) = \frac{\sigma}{r} \frac{1}{1 - 2\alpha \Pi' (Q_t)}.$$

As we discuss below, the above implies that prices exhibit excess, stochastic volatility, expected returns are predictable and can depend non-monotonically on Q_t, and serial correlation in returns is non-monotonic in horizon.

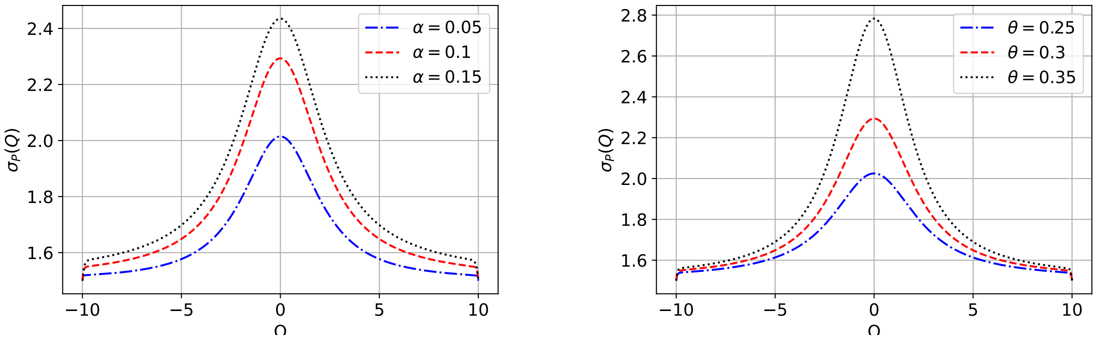
5.1 Stochastic volatility

Figure 5 provides an illustration of how stochastic volatility depends on parameters that capture the impact of Q-learners. In the absence of Q-learners, return volatility is constant and given by σ/r. However, in the presence of Q-learners, return volatility depends on the current state of Q_t, and is highest when Q_t is at zero. This captures the fact that, when $Q_t = 0$, the Q-learners are indifferent between buying and selling, which introduces maximal uncertainty and volatility in prices going forward. As a result, Q-learner demand $N_t^Q(\cdot)$ is most sensitive to changes in Q_t at this point. Moreover, the plots illustrate that return volatility is increasing in both the mass of Q-learners, θ, and their learning rate, α. These results are also intuitive and follow because trading by Q-learners induces an amplification effect. For instance, a positive innovation in dividends leads to higher prices directly, but

[31] It is worth emphasizing that this is the return that investors receive for holding one share of the risky asset, and therefore distinct from the rate of return one would receive from investing one dollar in the risky asset.

Figure 5: Return volatility with Q-learning

The figure plots return volatility $\sigma_P(Q)$ as a function of Q in the main equilibrium with hedging (Proposition 2), for different values of learning rate α and mass of Q-learners θ. Unless specified, parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $Y = 2$ and $m = 0.75$.



(a) Volatility $\sigma_P(Q)$ versus α

(b) Volatility $\sigma_P(Q)$ versus θ

also increases the net benefit Q_t of being long. This leads to a higher demand $N(Q_t)$ from Q-learners, which in turn, pushes the price up further. When the mass of Q-learners is larger, the variation in their trading behavior (as driven by the evolution of Q_t) has a larger impact on prices, and so increases volatility more. Similarly, when the learning rate α is higher, the net Q-value from buying is more volatile. This leads to more volatility in trading by Q-learners, and consequently, higher return volatility.

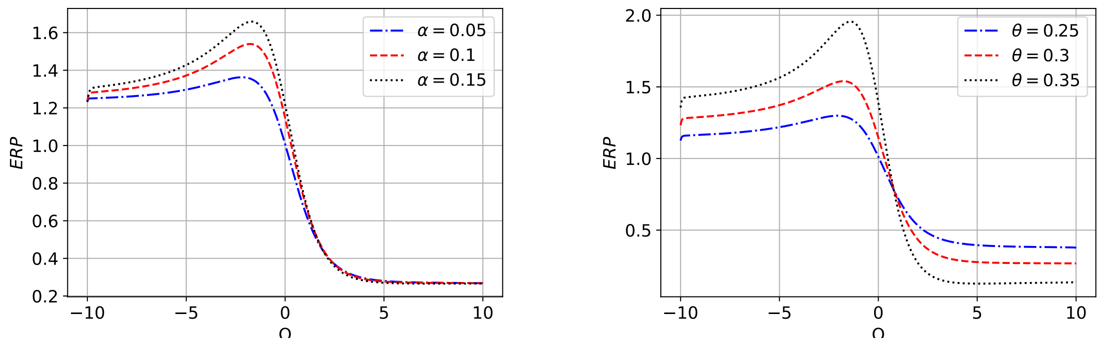
The model's prediction of stochastic volatility is consistent with empirical evidence on algorithmic trading. Using the introduction of co-location services as an exogenous instrument for algorithmic trading intensity, Boehmer et al. (2021) show that higher intensity of such trading *causally* increases short-term volatility across a number of empirical proxies and across 42 equity exchanges in 37 countries. Moreover, the prediction also distinguishes our setting from standard models of CARA investors with noise traders (e.g., Wang (1993)) or behavioral investors (e.g., Barberis et al. (2015)), which exhibit constant or deterministic return volatility.

5.2 Expected returns

Figure 6 illustrates the impact of Q-learning on the instantaneous expected return. Recall that in the absence of Q-learners, the instantaneous return is given by $\mathbb{E}[dR] = r\phi\sigma_P^2 S$. For the numerical illustration, we set the aggregate supply of the asset S to zero, which implies that in the absence of Q-learners, the expected return and Sharpe ratio are zero. In contrast, the expected return and Sharpe ratio vary with Q_t in the presence of Q-learners. It is worth

Figure 6: Expected return with Q-learning

The figure plots the instantaneous expected return $\mathbb{E}[dR_t]$ as a function of Q in the main equilibrium with hedging (Proposition 2), for different values of learning rate α and mass of Q-learners θ. Unless specified, parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $Y = 2$ and $m = 0.75$.



(a) Expected return $\mathbb{E}[dR_t]$ versus α

(b) Expected return $\mathbb{E}[dR_t]$ versus θ

noting that expected returns, and consequently, Sharpe ratios can be non-monotonic in Q. To see why, note that the impact of Q on the expected return can be decomposed as follows:

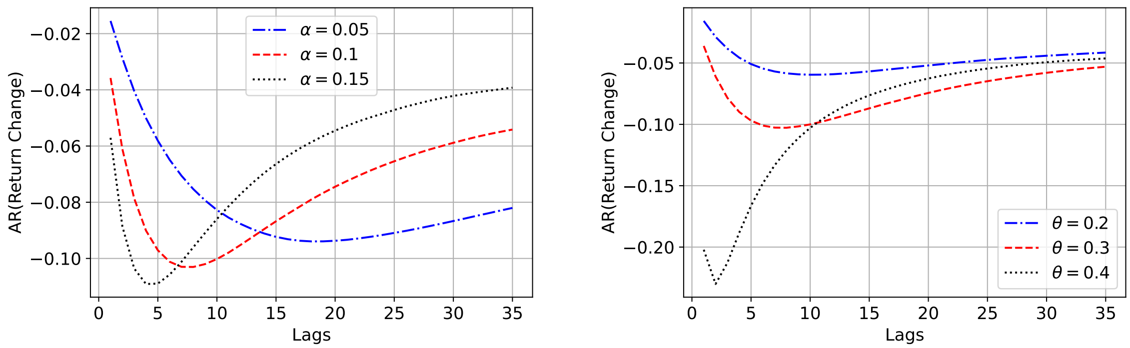
$$\mathbb{E}[dR_t] = \underbrace{\sigma_P^2(Q_t)}_{\text{scale effect}} \times \left(\underbrace{\frac{r\phi}{1-\theta}\left(S - \theta N^Q(Q_t)\right) + 2\alpha G'(Q_t)}_{\text{level effect}} \right).$$

Changes in Q_t affect expected returns through two channels. First, higher Q_t implies that Q-learners demand more of the asset, and so rational investors have to bear less risk in equilibrium, and require a lower risk-premium. This **level effect** implies that expected returns should be decreasing in Q_t. Second, as Figure 5 shows, stochastic volatility $\sigma_P^2(Q)$ rises as Q_t moves toward zero. This amplifies the level effect by increasing the *per-share* risk of holding the asset — we refer to this as the **scale effect**. It tends to make expected returns higher (lower) when Q_t is negative (positive) as Q_t approaches zero.

The overall impact of Q_t on expected returns depends on the interaction of the level and scale effects, as panels (a) and (b) of Figure 6 illustrate. When the scale effect is muted (e.g., when the rate of learning α or the mass of Q-learners θ is low), the level effect dominates and expected returns decrease with Q_t. However, when the scale effect is sufficiently large (e.g., for the dotted lines in either panel), the interaction with the scale effect implies that expected returns can be *increasing* in Q_t when Q_t is sufficiently away from zero in either direction. In other words, *higher* demand from Q-learners can lead to *higher* expected returns. This is because variation in demand from Q-learners affects not only the net supply of shares that

Figure 7: Autocorrelation in returns

The figure plots the autocorrelation in dollar returns $AC(\tau)$ at different lags τ in the main equilibrium with hedging (Proposition 2), for different values of learning rate α and mass of Q-learners θ. Unless specified, parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $Y = 2$ and $m = 0.75$.



(a) $AC(\tau)$ versus α

(b) $AC(\tau)$ versus θ

the rational investors have to hold in equilibrium, but also affects the risk per share (through its effect on stochastic volatility).

This non-monotonicity distinguishes our predictions from traditional models in which variation in demand from non-rational participants (e.g., noise traders) affects the net supply of shares that rational investors have to bear, but does not generate stochastic variation in the volatility of returns (e.g., Wang (1993), Wang (1994), Barberis et al. (2015)). Moreover, it is broadly consistent with the evidence in Brogaard et al. (2014) about the impact of algorithmic trading on future returns. Brogaard et al. (2014) find that liquidity demanding (marketable) trades by high-frequency traders are positively correlated with future returns, while liquidity supplying (non-marketable) trades are negatively correlated with such returns. While our model does not distinguish between marketable and non-marketable orders, one could interpret the Q-learners' demand from extreme Q_t as marketable, since they are less sensitive to Q_t, and consequently the price. In contrast, the demand at Q_t near zero is very sensitive to Q_t, and thus the price, and could be classified as non-marketable. Under this interpretation, the model predicts that an increase in the marketable demand (driven by an increase in Q_t in the extremes) is associated with an increase in expected returns, while an increase in non-marketable demand (driven by an increase in Q_t near zero) is associated with a decrease in expected returns.

5.3 Autocorrelation in returns

As Figure 7 illustrates, the mean-reverting properties of trading by Q-learners induce serial correlation in returns. Specifically, following Wang (1993), we plot the autocorrelation in returns at lag τ as:

$$AC(\tau) \equiv \frac{\mathbb{C}\left(R_{t+\tau} - R_t, R_t - R_{t-\tau}\right)}{\sqrt{\mathbb{V}\left(R_{t+\tau} - R_t\right)\mathbb{V}\left(R_t - R_{t-\tau}\right)}}.$$

The feedback into prices induced by Q-learners amplifies the negative correlation in the short-term, but mitigates it in the long-term. To gain some intuition, consider a period in which the dividend shock D_t is higher than expected and there is a contemporaneous increase in the price, P_t. This leads to a decrease in the net benefit from buying, Q_{t+dt}, over the next instant, which leads Q-learners to sell and prices to decrease going forward. Since Q_t is a persistent process, Q-learners continue to sell for a period, until the current price decreases sufficiently and Q_t starts increasing again.

All else equal, a higher learning rate α implies that Q-learners' trading is more sensitive to the realized return process. As panel (a) illustrates, this implies that the short-term serial correlation in returns is more negative, but also that it mean-reverts more quickly. In contrast, a larger mass θ of Q-learners leads to a larger negative impact in the short-term since the price impact of a unit change in Q_t is larger. However, the autocorrelation at longer lags is approximately the same for different levels of θ.

The patterns in autocorrelation are broadly consistent with empirical evidence about variation in serial correlation across different horizons (e.g., Heston, Korajczyk, and Sadka (2010), Bogousslavsky (2016), Baule, Schlie, and Zhou (2025)). Using high-frequency data on US stocks, Baule et al. (2025) document a term-structure of return autocorrelation that is qualitatively similar to our model predictions. They show that average intraday return autocorrelations are negative across most horizons, are close to zero for sub-minute returns, decline to a minimum for intermediate horizons (e.g., around 15 minutes) and then gradually revert to zero for longer horizons.

Our model generates novel testable predictions that relate the term structure of return autocorrelation to the intensity and speed of algorithmic trading. To the extent that such traders have a larger impact (higher θ) and learn more slowly (lower α) for small stocks, our model predictions are consistent with the additional evidence from Baule et al. (2025) that suggests small stocks exhibit a sharper drop and slower reversion in return autocorrelation.

6 Welfare, risk sharing, and policy

This section uses the equilibrium of Section 4.2 to evaluate how the presence of Q-learners affects investor welfare. We show that Q-learners learn to provide liquidity to one side of the market, which allows them to realize profits. While investors on the same side as Q-learners lose, since they now face additional competition, investors on the opposite side gain as they benefit from more favorable prices. In aggregate, the presence of Q-learners can improve the average utility across rational investors, even though it leads to higher volatility and decreases utility for some fraction of the investors.

Again, we focus throughout on the case $\bar{Y} = (2m-1)Y > 0$ (equivalently $m > 1/2$, $Y > 0$), so that the rational sector's aggregate hedging demand is to short the asset; the case $\bar{Y} < 0$ is symmetric.

6.1 Q-learner profits and investor welfare

Figure 8 provides an illustration. Specifically, the figure plots the expected discounted profits $U(Q_t)$ for a Q-learner, where

$$U(Q_t) = E\left[\int_t^\infty e^{-r(s-t)} N^Q(Q_s)\, dR_s\right],$$

and the utility gains $G_S(Q_t)$ and $G_L(Q_t)$ for S and L investors.[32]

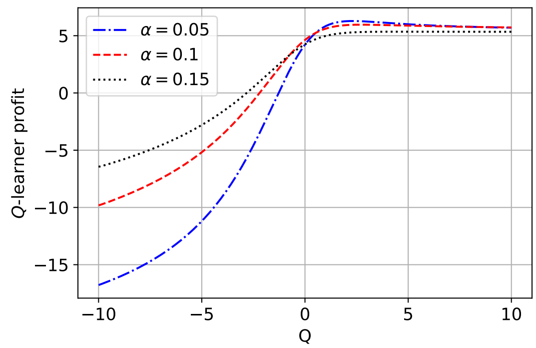
Because the rational investors' net hedging exposure is $\bar{Y} > 0$, S-investors collectively want to short the asset more than L-investors want to go long, and the equilibrium price clears at a discount. As a result, as panel (a) illustrates, the expected profit for a Q-learner can be positive when she is sufficiently long (i.e., when Q_t is sufficiently above zero), since she is more likely to buy the security when its price is low. In contrast, when Q_t is negative, the Q-learner is short the security and, consequently, her expected profit is negative. Over time, the algorithm "learns" that providing liquidity to S investors by buying shares is profitable in equilibrium.

A key feature of the equilibrium is that rational investors are *not* uniformly better off in the presence of Q-learners. In the no-hedging benchmark ($Y = 0$, Appendix A.1), rational investors earn trading profits at the expense of Q-learners and so weakly gain from their presence. With $\bar{Y} > 0$, however, the two types are differentially affected: S investors benefit because Q-learners absorb part of their risk, while L-investors lose because Q-learners compete away part of their risk-sharing profits. Whether Q-learners help or hurt a given rational investor therefore depends on which side of the hedging market she is on. Panels (b)–(c) of

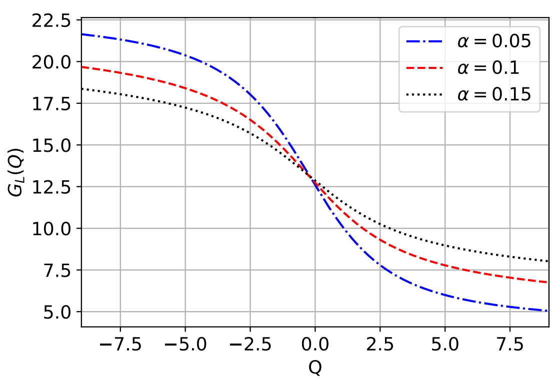
[32] While the Q-learner does not optimize using a discount rate r, we use it as a welfare benchmark.

Figure 8: Q-learner profits and rational investor utility gains

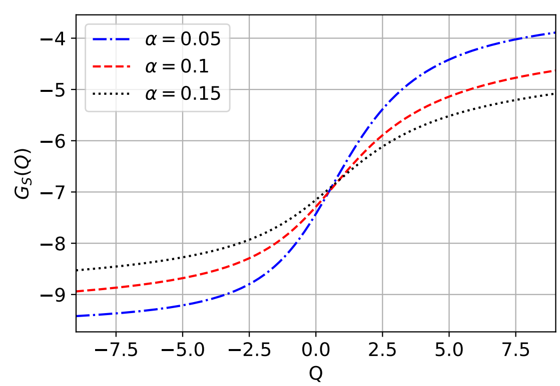
The figure plots the expected discounted profits for a Q-learner, $U(Q_t)$, and the utility gains for L and S investors, as functions of Q, for different values of learning rate α. Other parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $m = 0.75$ and $Y = 2$.



(a) Q-learner profits versus α



(b) Long utility gain $G_L(Q)$ versus α



(c) Short utility gain $G_S(Q)$ versus α

Figure 8 make this concrete: $G_L(Q)$ shifts down and $G_S(Q)$ shifts up as the learning rate α rises.

The distribution of welfare changes across L and S is also asymmetric in magnitude: because $m > 1/2$, the gain to S-investors is weighted more heavily in the cross-sectional average than the loss to the L-investors, which is why the average rational welfare can increase in the presence of Q-learners even though some investors are strictly worse off. Section 6.2 traces out the comparative statics behind this aggregation.

6.2 Comparative statics

Figure 9 characterizes how expected Q-learner profits and rational traders' utility gains depend on model parameters. Specifically, the figure plots expected discounted profits (i.e.,

U), utility gains for L and S traders (i.e., $G_L \equiv \mathbb{E}\left[G_L\left(Q_t\right)\right]$ and $G_S \equiv \mathbb{E}\left[G_S\left(Q_t\right)\right]$, respectively) and the average utility gain (i.e., $\bar{G} = mG_S + (1-m)\,G_L$), where the expectations are computed using the steady state distribution of Q_t.[33] Utility gains for rational investors depend on (i) profits generated by trading against Q-learners (as in the main model) and (ii) hedging needs due to risky endowments. As panel (a) illustrates, when Y is small, trading profits dominate and so average G_L, G_S and $\bar{G}$ are positive while average U is negative. However, as Y increases, hedging demands begin to dominate. As a result, Q-learner profits are increasing in Y (all else equal) and become positive for sufficiently high Y. Moreover, since aggregate hedging demand from S investors is larger, G_S decreases with Y while G_L increases with Y.

Panel (b) illustrates the impact of varying the mass m of S investors on trading profits and utility gains. When m is low, hedging demand from rational investors is dominated by L investors (i.e., $\bar{Y} < 0$), and consequently, trading gains for S investors and Q-learners are high. Similarly, when m is sufficiently high, hedging demand from S investors dominates and so trading gains for L investors and Q-learners is high. For intermediate m, the demand from S and L investors offset each other, so the net demand from rational investors is not very large. In this case, Q-learners do not earn large profits from providing risk sharing, but instead are a source of trading profits for rational investors. As a result, Q-learner profits are negative, while average utility gains for rational investors are higher than for extreme m.

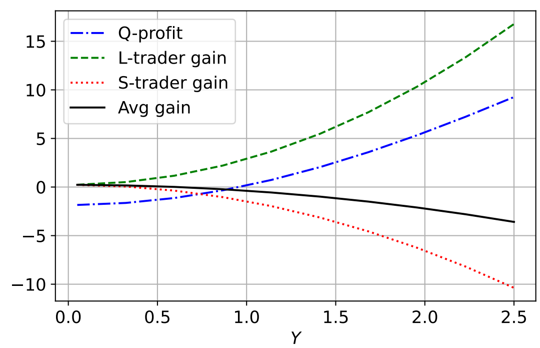
Intuitively, as the mass of Q-learners increases, average utility gains for rational investors increase while trading profits for Q-learners decrease — panel (c) provides an illustration of this. Recall that in this illustration, $m > 1/2$ and rational hedging demand is dominated by S investors. An increase in θ implies there are more Q-learners to provide risk sharing, which improves G_S but reduces G_L. Moreover, an increase in θ leads to higher trading profits for rational investors, which tends to increase the average utility gain.

In contrast, panel (d) shows that an increase in fundamental volatility σ can have a non-monotonic impact on Q-learner profits. This is because an increase in volatility has two effects: (i) it increases the benefit from providing risk sharing to rational investors, and (ii) it leads to larger expected losses from trading against better informed investors. When σ is low, the first effect dominates and Q-learner profits increase with volatility. However, when σ is sufficiently large, the second effect dominates and profits can decrease with volatility. Moreover, in this case, average utility gains for rational investors also increase.

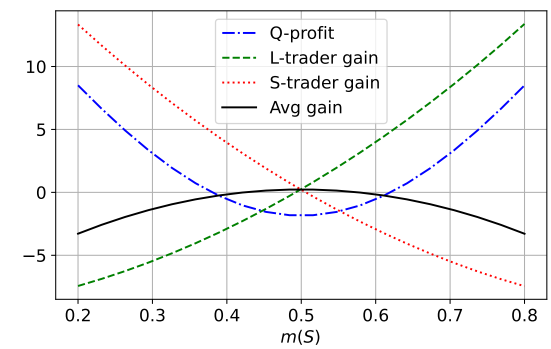
[33] It is worth noting that $\bar{G}$ does not correspond to the utility gain of rational investors on average because of Jensen's inequality. In Figure 14 of Appendix F, we present the corresponding plots for the expected value function for L and S traders (i.e., $\mathbb{E}[V_L(Q_t)]$ and $\mathbb{E}[V_S(Q_t)]$) and the weighted average $m\mathbb{E}[V_S(Q_t)] + (1-m)\,\mathbb{E}[V_L(Q_t)]$, where the expectations are taken under the steady state distribution. While the comparative statics are qualitatively similar, it is clearer to see the impact of parameters when we plot expected utility gains $G(\cdot)$, and so we do so in the main text.

Figure 9: Expected trading profits and utility gains

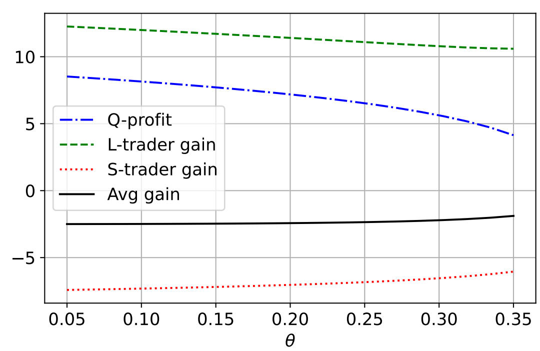
The figure plots the expected discounted profits for a Q-learner, $U(Q_t)$ (dot-dashed), and the utility gains for L and S investors, G_L (dashed) and G_S (dotted), and the average gain $\bar{G} = mG_S + (1-m)\,G_L$ (solid), versus various model parameters. Unless specified, parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $m = 0.75$ and $Y = 2$.



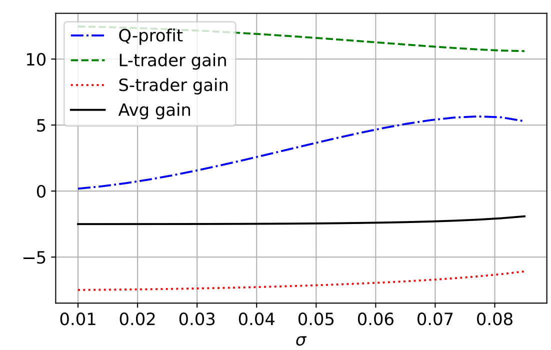
(a) Average Profits / utility gains versus Y



(b) Average Profits / utility gains versus m



(c) Average Profits / utility gains versus θ



(d) Average Profits / utility gains versus σ

6.3 Policy implications

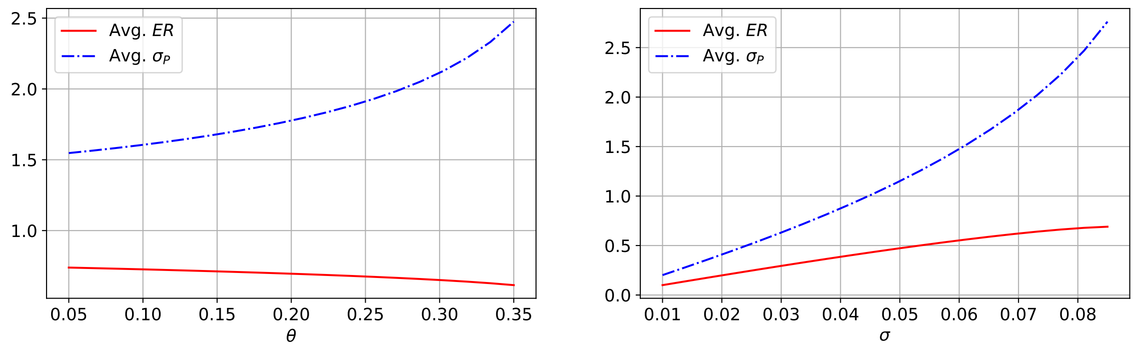
Regulators worry that algorithmic trading raises volatility and so harms investor welfare. The Bank of England has warned that algorithmic strategies could "cause unanticipated market disruptions" and the OECD has flagged the risk that high-frequency algorithms "increase market instability and harm ordinary investors."[34]

Our results suggest this concern conflates two distinct things: return volatility and investor welfare. In our equilibrium both move in the *same* direction when participation by Q-learners (θ) rises. Figure 10 provides an illustration. For the same parameters considered in Figure 9, panel (a) implies that an increase in the fraction of Q-learners leads to higher

[34]See Bank of England, Financial Policy Committee (April 2025) and OECD, Regulatory approaches to Artificial Intelligence in finance (2024).

Figure 10: Expected returns and average return volatility

The figure plots the expected return and average return volatility versus various model parameters. Unless specified, parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $m = 0.75$ and $Y = 2$.



(a) Average ER and σ_P versus θ

(b) Average ER and σ_P versus σ

return volatility due to the feedback in trading they generate. Similarly, panel (b) suggests that the presence of Q-learners leads to a sharp amplification effect: return volatility increases nonlinearly in cash-flow volatility. However, panels (c) and (d) of Figure 9 imply that average trading gains for rational investors increase with θ and σ, respectively, for these parameter values. Intuitively, since rational investors are risk averse and have hedging needs, while Q-learners are risk-neutral, there are gains from trade. Moreover, since rational investors have an informational advantage relative to Q-learners, an increase in the presence of Q-learners can lead to higher welfare due to more trading profits, despite the increase in return volatility. Treating higher volatility as evidence of welfare harm therefore mismeasures the welfare consequences of algorithmic trading, and restricting algorithmic participation in this setting would reduce risk-sharing capacity, not improve investor welfare.

7 Conclusions

We develop a model in which rational investors and Q-learners trade a risky security. The trading behavior of the Q-learners is driven by their perceived net benefit from buying a share, Q_t, which we show can be approximated in continuous-time by an endogenous SDE. This allows us to analytically characterize equilibrium prices and trading. The Q-learners' trading generates a feedback loop in equilibrium prices, which leads to stochastic volatility, state-dependence in expected returns, and novel patterns in return autocorrelation which depend on the mass and learning rate of Q-learners. Our model also allows us to characterize the impact of Q-learners on investor utility. We show that when risk sharing is an important trading motive for investors, Q-learners can earn positive profits and improve average investor utility, even though they increase the volatility of prices. This has direct policy implications for the regulation of algorithmic trading.

Our model is stylized for analytical tractability and expositional clarity. There are a number of natural directions for future work. First, it would be interesting to explore human-AI interactions in other market environments (e.g., in strategic trading settings). Second, we have assumed that the Q-learners do not have an informational advantage over the rational investors. However, in practice, one argument in favor of algorithmic trading is that such approaches allow the use of better information (e.g., in the form of big data). Incorporating both superior information and a lack of sophistication (e.g., about the structure of the economy) in Q-learning traders would be another interesting avenue.

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A Proofs

A.1 The no-liquidity-demand benchmark ($Y = 0$)

In the main text we work with the general model in which rational investors have heterogeneous hedging needs. Here we first record the nested *no-liquidity-demand benchmark* with $Y = 0$, in which both rational types are identical ($\bar{Y} = 0$) and trade purely for speculative reasons. This benchmark isolates the Q-feedback channel from the risk-sharing channel and gives the simplest characterization of how Q-learners reshape the equilibrium. Proposition A.1 below characterizes it, and its proof underlies the existence argument for our main result, Proposition 2.

With $Y = 0$ the value function of the (representative) rational investor is

$$V(W_t, Q_t) = -\frac{1}{r\phi} \exp\left(-\phi r W_t - \frac{\delta - r}{r} - G(Q_t)\right), \tag{20}$$

where $G(Q_t)$ is the utility gain from being able to predict the Q-learners' demand. The first-order conditions imply optimal consumption

$$C_t = rW_t + \frac{\delta - r}{r\phi} + \frac{1}{\phi} G(Q_t) \tag{21}$$

and optimal demand

$$N_t^R = \underbrace{\frac{D_t + \mu_P(Q_t) - rP_t}{\phi r \sigma_P^2(Q_t)}}_{\text{speculative}} - \underbrace{2\alpha \frac{G'(Q_t)}{\phi r}}_{Q\text{-anticipation}}, \tag{22}$$

and market clearing pins down the price

$$P_t = \frac{1}{r}\left(D_t + \mu_P(Q_t) - \sigma_P^2(Q_t)\left(\frac{r\phi}{1-\theta}\left(S - \theta N^Q(Q_t)\right) + 2\alpha G'(Q_t)\right)\right). \tag{23}$$

The following result characterizes existence of the benchmark Q-REE.

Proposition A.1. *There exists a Q-REE where the rational trader's value function is given by (20), optimal consumption is given by (21), optimal demand is given by (22), and the equilibrium price is given by (5) where the risk premium $\Pi(Q)$ satisfies the nonlinear*[35] *second-order ODE*

$$\begin{aligned} r\Pi(Q) = {} & \Pi'(Q)\mu_Q(Q) + \frac{1}{2}\Pi''(Q)\sigma_Q^2(Q) \\ & - G'(Q)\sigma_Q(Q)\sigma_P(Q) - \frac{r\phi\sigma_P^2(Q)}{1-\theta}\left(S - \theta N^Q(Q)\right) \end{aligned} \tag{24}$$

on $\mathbb{R}$ with boundary conditions $\Pi'(-\infty) = \Pi'(\infty) = 0$, and for all Q, $\Pi'(Q) < \frac{1}{2\alpha}$. The

[35] The coefficients $\mu_P(\cdot)$, $\mu_Q(\cdot)$, $\sigma_P(\cdot)$, and $\sigma_Q(\cdot)$ depend on the function $\Pi(\cdot)$, see Equations (17) and (16).

utility gain $G(Q)$ satisfies the nonlinear second-order ODE

$$rG(Q) = \frac{1}{2}\sigma_P^2(Q)\left(\frac{r\phi}{1-\theta}\left(S - \theta N^Q(Q)\right)\right)^2 + G'(Q)\mu_Q(Q) \tag{25}$$
$$+\left(G''(Q) - G'(Q)^2\right)\frac{1}{2}\sigma_Q^2(Q)$$

on $\mathbb{R}$ with boundary conditions $G'(-\infty) = G'(\infty) = 0$. The evolution of Q is driven by the drift and diffusion terms:

$$\mu_Q(Q) = \frac{2\alpha}{1-2\alpha\Pi'(Q)}\left(\tfrac{1}{2}\sigma_Q^2(Q)\Pi''(Q) - r\Pi(Q) - \tfrac{1}{2}Q\right) \quad \textit{and} \quad \sigma_Q(Q) = \frac{\sigma}{r}\frac{2\alpha}{1-2\alpha\Pi'(Q)}.$$

The expressions in (24) and (25) can be derived by (i) plugging in the expressions for μ_P and σ_P from (17) into (23), and (ii) plugging in the optimal consumption and demand expressions and the market clearing condition into the trader's HJB equation (26) and simplifying. We detail the argument below.

A.1.1 Traders' HJB Equation

Given the SDEs (2), (14), and (15), standard verification arguments imply that the solution to the trader's optimization problem $V(W,Q)$ in Equation (6) is characterized by the following Hamilton-Jacobi-Bellman (HJB) equation:

$$\begin{aligned}\delta V &= \max_{C,N} -\frac{1}{\phi}e^{-\phi C} + V_W\left(rW - C + N\left(\mu_P(Q) - rP(D,Q) + D\right)\right)\\ &\quad + \frac{1}{2}V_{WW}\sigma_P^2(Q)N^2\\ &\quad + V_Q\mu_Q(Q) + V_{QQ}\frac{1}{2}\sigma_Q^2(Q)\\ &\quad + V_{WQ}N\sigma_P(Q)\sigma_Q(Q),\end{aligned}$$

where we omit the dependence of V on (W,Q) to keep the notation simple. Conjecture that $V(W,Q) = -\exp(-\phi rW - G(Q) - K)$, where

$$K = \frac{\delta - r}{r} + \log(\phi r). \tag{27}$$

The first-order condition for N is given by

$$N = -\frac{V_W}{V_{WW}}\frac{\mu_P(Q) - \frac{\mu}{r} - r\Pi(Q)}{\sigma_P^2(Q)} - \frac{V_{WQ}}{V_{WW}}\frac{\sigma_Q(Q)}{\sigma_P(Q)}. \tag{28}$$

The conjecture implies that $V_{WQ} = \phi rG'(Q)V$, $V_W = -\phi rV$, and $V_{WW} = (\phi r)^2 V$, so that we can simplify this expression to

$$N = \frac{1}{\phi r}\frac{\mu_P(Q) - \frac{\mu}{r} - r\Pi(Q)}{\sigma_P^2(Q)} - \frac{G'(Q)}{\phi r}\frac{\sigma_Q(Q)}{\sigma_P(Q)}.$$

Plugging in $\sigma_Q(Q) = 2\alpha\sigma_P(Q)$ yields Equation (22).

The market clearing condition (7) implies that

$$N = \frac{1}{1-\theta}\left(S - \theta N^Q(Q)\right)$$

or equivalently

$$\frac{1}{\phi r}\frac{\mu_P(Q) - \frac{\mu}{r} - r\Pi(Q)}{\sigma_P^2(Q)} - 2\alpha\frac{G'(Q)}{\phi r} = \frac{1}{1-\theta}\left(S - \theta N^Q(Q)\right).$$

After some algebra, this implies

$$\begin{aligned} r\Pi(Q) &= \Pi'(Q)\mu_Q(Q) + \frac{1}{2}\Pi''(Q)\sigma_Q^2(Q) \\ &\quad - G'(Q)\sigma_Q(Q)\sigma_P(Q) - \frac{r\phi\sigma_P^2(Q)}{1-\theta}\left(S - \theta N^Q(Q)\right), \end{aligned}$$

which is Equation (24).

The FOC for consumption is given by

$$C = -\frac{1}{\phi}\log(V_W),$$

which yields Equation (21). To derive $G(Q)$, plug in the FOC for N in Equation (28) and the FOC for C into the HJB Equation (26) and use $V_{QQ} = \left(G'(Q)^2 - G''(Q)\right)V$, which yields

$$\begin{aligned} \delta V &= rV - \phi r V\left(\frac{1}{\phi}\log(\phi r) - \frac{1}{\phi}G(Q) - \frac{1}{\phi}K\right) \\ &\quad + \frac{1}{2}(\phi r)^2 V\sigma_P^2(Q) N^2 \\ &\quad - G'(Q)V\mu_Q(Q) + \left(G''(Q) - G'(Q)^2\right)V\frac{1}{2}\sigma_Q^2(Q). \end{aligned}$$

Canceling V and plugging in the market clearing Condition (7) for N yields

$$\begin{aligned} \delta &= r - r\log(\phi r) + rG(Q) + rK \\ &\quad - \frac{1}{2}(\phi r)^2\sigma_P^2(Q)\left(\frac{1}{1-\theta}\left(S - \theta N^Q(Q)\right)\right)^2 \\ &\quad - G'(Q)\mu_Q(Q) + \left(G'(Q)^2 - G''(Q)\right)\frac{1}{2}\sigma_Q^2(Q). \end{aligned}$$

$$\begin{aligned} rG(Q) &= \frac{1}{2}\sigma_P^2(Q)\left(\frac{r\phi}{1-\theta}\left(S - \theta N^Q(Q)\right)\right)^2 + G'(Q)\mu_Q(Q) \\ &\quad + \left(G''(Q) - G'(Q)^2\right)\frac{1}{2}\sigma_Q^2(Q) \end{aligned}$$

Plugging in the expression for K in Equation (27) and rearranging the expressions involving $G(Q)$ yields Equation (25).

A.1.2 Existence of Solutions for Equations (24) and (25)

The proof proceeds as follows. We first fix $G'(Q)$ to be an arbitrary continuous function which is bounded on $\mathbb{R}$ and we show that, taking $G'(Q)$ as given, Equation (24) has a twice continuously differentiable and bounded solution on $\mathbb{R}$, which satisfies $\Pi'(Q) \in (-R, 1/(2\alpha))$ for some $R > 0$. Then, we show that for any continuous $\Pi'(Q)$ such that $\Pi'(Q) \in (-R, 1/(2\alpha))$, Equation (25) has a twice continuously differentiable and bounded solution on $\mathbb{R}$.

Taken together, these two results imply that we can define a modification of the ODE system (24) and (25), which is bounded in both the function values $\Pi(Q)$ and $G(Q)$ and the derivatives $\Pi'(Q)$ and $G'(Q)$. Any solution of that ODE system is also a solution of the original system (24) and (25). We then prove that the modified system has a solution on any bounded interval of Q, and then use the Arzela-Ascoli theorem to extend the solutions to the entire real line.

Proposition A.2. *Fix $G'(Q)$ to be an arbitrary continuous function, which satisfies $\sup_{Q\in\mathbb{R}} |G'(Q)| < \infty$ and $\sup_{Q\in\mathbb{R}} |G''(Q)| < \infty$. Then, Equation (24) has at least one solution $\Pi(Q) \in \mathcal{C}^2(\mathbb{R})$.*[36] *Any solution satisfies $\sup_{Q\in\mathbb{R}} |\Pi(Q)| < \infty$ and $\Pi'(Q) \in (-R, 1/(2\alpha))$ for all $Q \in \mathbb{R}$ for some $R > 0$.*

Proof. Rewrite Equation (24) as

$$\Pi''(Q) = F_\Pi\left(Q, \Pi(Q), \Pi'(Q), G'(Q)\right) \tag{30}$$

where

$$F_\Pi\left(Q, u, v, G'(Q)\right) = (ru + \alpha Q v)\, 2\frac{(1-2\alpha v)^2}{4\alpha^2\left(\frac{\sigma}{r}\right)^2} + \frac{1}{2\alpha^2}(1-2\alpha v)\left(2\alpha G'(Q) + \frac{r\phi}{1-\theta}\left(S - \theta N^Q(Q)\right)\right).$$

Pick $M > 0$ sufficiently large and define for $(Q, u, v) \in \mathbb{R}^3$

$$\bar{F}_\Pi\left(Q, u, v, G'(Q)\right) = F_\Pi\left(Q, u, \max\{\min\{v, M\}, -M\}, G'(Q)\right).$$

Consider the ODE

$$\bar{\Pi}''(Q) = \bar{F}_\Pi\left(Q, \bar{\Pi}(Q), \bar{\Pi}'(Q), G'(Q)\right). \tag{31}$$

We now apply De Coster and Habets (2006), Th. 5.6, p. 122, to Equation (31), which is restated below for the convenience of the reader.

Theorem 1 (De Coster and Habets (2006), Th. 5.6)**.** *Let $A(Q), B(Q) \in \mathcal{C}^2(\mathbb{R})$ such that $A(Q) \le B(Q)$ for all $Q \in \mathbb{R}$, $E = \{(Q, u, v) \in \mathbb{R}^3 | A(Q) \le u \le B(Q)\}$ and let $f : E \to \mathbb{R}$ be continuous.*

Assume that $A(Q)$ and $B(Q)$ are such that for all $Q \in \mathbb{R}$,

$$A''(Q) \ge f\left(Q, A(Q), A'(Q)\right) \text{ and } B''(Q) \le f\left(Q, B(Q), B'(Q)\right).$$

[36] Here, $\mathcal{C}^2(\mathbb{R})$ is the space of twice continuously differentiable functions on $\mathbb{R}$.

Also, assume that for any bounded interval I, there exists a positive continuous function $\varphi_I : \mathbb{R}^+ \to \mathbb{R}$ that satisfies

$$\int_0^\infty \frac{sds}{\varphi_I(s)} = \infty \tag{32}$$

and for all $Q \in I$, $(u,v) \in \mathbb{R}^2$ with $A(Q) \le u \le B(Q)$,

$$|f(Q,u,v)| \le \varphi_I(|v|).$$

Then, the ODE

$$u''(Q) = f(Q, u(Q), u'(Q))$$

has at least one solution $u \in \mathcal{C}^2(\mathbb{R})$ such that for all $Q \in \mathbb{R}$

$$A(Q) \le u(Q) \le B(Q).$$

In particular, we show that there exist numbers $A < B$ such that the constant functions $A(Q) = A$ and $B(Q) = B$ are sub-solutions and super-solutions to Equation (31), respectively, on $\mathbb{R}$ and that $\bar{F}_\Pi(Q,u,v,G'(Q))$ satisfies all conditions of the theorem.

We have

$$F_\Pi(Q,A,0,G'(Q)) = \frac{rA}{2\alpha^2\left(\frac{\sigma}{r}\right)^2} + \frac{1}{2\alpha^2}\left(2\alpha G'(Q) + \frac{r\phi}{1-\theta}\left(S - \theta N^Q(Q)\right)\right).$$

Since $G'(Q)$ and $N^Q(Q)$ are bounded on $\mathbb{R}$, we have $F_\Pi(Q,A,0,G'(Q)) < 0$ all $Q \in \mathbb{R}$ whenever A is sufficiently small. In particular, it holds that $A''(Q) = 0 \ge F_\Pi(Q,A(Q),A'(Q),G'(Q)) = F_\Pi(Q,A,0,G'(Q))$, so that $A(Q) = A$ is indeed a subsolution to Equation (30). Since $A'(Q) = 0 \in (-R, 1/(2\alpha))$, it is also a subsolution to Equation (31). Similarly, for B sufficiently large, $F_\Pi(Q,B,0,G'(Q)) > 0$ for all $Q \in \mathbb{R}$ and by the same argument, $B(Q) = B$ is a supersolution to Equations (30) and (31). Since $F_\Pi(Q,u,0,G'(Q))$ is strictly increasing in u, and $F_\Pi(Q,A,0,G'(Q)) < F_\Pi(Q,B,0,G'(Q))$, it must be the case that $B > A$.

For any bounded interval $I \subset \mathbb{R}$, define $\varphi_I(v) = \max_{(Q,u,v)\in I\times[A,B]\times\mathbb{R}} \left|\bar{F}_\Pi(Q,u,v,G'(Q))\right|$ and note that $\varphi_I(v)$ satisfies the Nagumo condition in Equation (32). By construction of $\varphi_I(v)$, we have $\left|\bar{F}_\Pi(Q,u,v,G'(Q))\right| \le \varphi_I(|v|)$ for any $(Q,u,v) \in I \times [A,B] \times \mathbb{R}$.

Hence, the theorem implies that Equation (31) has at least one solution $\bar{\Pi}(Q)$, which is twice continuously differentiable on $\mathbb{R}$ and which satisfies $\bar{\Pi}(Q) \in [A,B]$ for all $Q \in \mathbb{R}$. □

We next show that any solution of Equation (31) has a bounded derivative. This implies that for M sufficiently large, any solution to Equation (31) is also a solution to Equation (30).

Lemma 1. *Any solution $\bar{\Pi}(Q)$ to Equation (31) satisfies $\bar{\Pi}'(Q) \in (-R, 1/(2\alpha))$ for some $R > 0$ sufficiently large.*

Proof. We first show that $\bar{\Pi}'(Q) < 1/(2\alpha)$. Pick $M > 1/(2\alpha)$. By way of contradiction, suppose that there exists a solution $\bar{\Pi}(Q)$ to Equation (31) such that $\bar{\Pi}'(Q) = 1/(2\alpha)$ for some Q. Then, since $\bar{\Pi}(Q)$ is bounded, Equation (31) implies that $\bar{\Pi}''(Q) = 0$. Hence, $\bar{\Pi}'(Q') = 1/(2\alpha)$ for $Q' > Q$. But this implies that $\bar{\Pi}(Q) > B$ for Q sufficiently large, which is a contradiction. Similarly, if there exists a Q such that $\bar{\Pi}'(Q) > 1/(2\alpha)$, it must be the case that for some $Q' > Q$, $\bar{\Pi}'(Q') = 1/(2\alpha)$, otherwise $\bar{\Pi}(Q) \to \infty$ as $Q \to \infty$, which contradicts the fact that $\bar{\Pi}(Q) \in [A, B]$. But then, the same contradiction holds at Q'. Thus, any solution to Equation (31) must satisfy $\bar{\Pi}'(Q) < 1/(2\alpha)$ for all $Q \in \mathbb{R}$.

Next, we show that $\bar{\Pi}'(Q) > -R$ for some $R > 0$ sufficiently large. Wlog pick R so that

$$(1 + 2\alpha R)(r + \alpha) R \frac{1}{2\alpha^2 \left(\frac{\sigma}{r}\right)^2} > \frac{1}{2\alpha^2}\left(2\alpha G''(Q) + \frac{r\phi}{1-\theta}\left(S - \theta N^{Q'}(Q)\right)\right) \tag{33}$$

for all Q, which is possible because $G''(Q)$ and $N^{Q'}(Q)$ are bounded, and pick $M > R$. Suppose by way of contradiction that for some Q, $\bar{\Pi}'(Q) \leq -R$. Then, since $\bar{\Pi}(Q)$ is bounded, it must hold that $\lim_{|Q|\to\infty} \bar{\Pi}'(Q) = 0$ and there exists a $\hat{Q} = \inf\left\{Q : \bar{\Pi}'(Q) \leq -R\right\}$ for which

$$\begin{aligned}\bar{\Pi}''\left(\hat{Q}\right) = F_{\Pi}\left(\hat{Q}, \bar{\Pi}\left(\hat{Q}\right), R, G'\left(\hat{Q}\right)\right) &= (1 + 2\alpha R)\Bigg(\left(r\bar{\Pi}\left(\hat{Q}\right) - \alpha \hat{Q} R\right)\frac{1}{2\alpha^2\left(\frac{\sigma}{r}\right)^2}(1 + \alpha R) \\ &\quad + \frac{1}{2\alpha^2}\left(2\alpha G'\left(\hat{Q}\right) + \frac{r\phi}{1-\theta}\left(S - \theta N^{Q}\left(\hat{Q}\right)\right)\right)\Bigg) \\ &< 0.\end{aligned}$$

That is, $\hat{Q}$ is the first time $\bar{\Pi}'(Q)$ crosses $-R$ from above. Since we picked R sufficiently large, Equation (33) implies that the RHS is strictly decreasing in Q whenever $\bar{\Pi}'(Q) \leq -R$. But this implies that $\bar{\Pi}''(Q') < 0$ for all $Q' > Q$ if $\bar{\Pi}'(Q') = -R$. Therefore $\bar{\Pi}'(Q') \leq -R$ for all $Q' > Q$, which contradicts the fact that $\bar{\Pi}'(\infty) = 0$. Hence, $\bar{\Pi}'(Q) > -R$ for all Q. □

Lemma 1 immediately implies the following.

Corollary 2. *For R sufficiently large, any solution of Equation (31) also satisfies Equation (30).*

The corollary implies that Equation (30) has at least one solution $\Pi(Q)$ such that $\Pi(Q) \in [A, B]$ and $\Pi'(Q) \in (-R, 1/(2\alpha))$, which is what we set out to prove.

Next, we show that for any function $\Pi(Q)$ such that $\Pi'(Q)$ is continuous and $\Pi'(Q) \in (-R, 1/(2\alpha))$ for all $Q \in \mathbb{R}$, Equation (25) has a solution, such that $G'(Q)$ is continuous and bounded on $\mathbb{R}$.

Proposition A.3. *Fix $\Pi'(Q)$ to be an arbitrary continuous function such that for some $R > 0$, $\Pi'(Q) \in (-R, 1/(2\alpha))$ for all $Q \in \mathbb{R}$. Then Equation (25) has a bounded solution $G(Q) \in \mathcal{C}^2(\mathbb{R})$ and $G'(Q)$ is bounded on any finite interval $I \subset \mathbb{R}$.*

Proof. For a given $\Pi'(Q)$, define

$$\sigma_P(Q) = \frac{\sigma}{r}\frac{1}{1 - 2\alpha\Pi'(Q)}$$

and notice that $\sigma_P(Q)$ continuous and uniformly bounded on $\mathbb{R}$,[37] and satisfies $\inf_{Q\in\mathbb{R}}\sigma_P(Q) \geq$

[37] This follows because for any Q, $\Pi'(Q) < 1/(2\alpha)$ and $\Pi'(Q) \to 0$ as $|Q| \to \infty$, so $\Pi'(Q)$ cannot approach $1/(2\alpha)$ as $|Q|$ approaches infinity.

$\underline{\sigma} > 0$ for some $\underline{\sigma} > 0$.[38] Proceeding similarly as for Equation (24), write Equation (25) as

$$G''(Q) = F_G(Q, G(Q), G'(Q), \sigma_P(Q)) \tag{34}$$

where

$$\begin{aligned} F_G(Q, u, v, \sigma_P(Q)) &= (ru + \alpha Q v) \frac{1}{2\alpha^2 \sigma_P^2(Q)} \\ &\quad - \frac{1}{4\alpha^2} \left(\frac{r\phi}{1-\theta} \left(S - \theta N^Q(Q) \right) + 2\alpha v \right)^2. \end{aligned}$$

We now apply the same theorem as above, De Coster and Habets (2006), Th. 5.7, p. 122, to Equation (34). Since $\Pi'(Q)$ and $N^Q(Q)$ are bounded and since $F_G(Q, u, v, \sigma_P(Q))$ is increasing in u, there exist $A < B$ such that the constant functions $A(Q)$ and $B(Q)$ are sub-solutions and super-solutions to Equation (34). For any bounded interval $I \subset \mathbb{R}$, pick

$$\varphi_I(v) = K_I \left(1 + v^2\right)$$

for some $K_I > 0$. Then, $\varphi_I(v)$ satisfies Equation (32) and

$$|F_G(Q, u, v, \sigma_P(Q))| \leq \varphi_I(|v|) \tag{35}$$

whenever K_I is sufficiently large. Hence, Equation (34) has at least one solution $G(Q) \in \mathcal{C}^2(\mathbb{R})$ which satisfies $G(Q) \in [A, B]$ for all $Q \in \mathbb{R}$.

It remains to show that any solution of Equation (34) has a derivative $G'(Q)$ which is bounded on finite intervals. This follows from Hartman (2002), Lemma 5.1, p. 428. The lemma states that if G is bounded on $\mathbb{R}$ and $G''(Q) \leq \varphi(|G'(Q)|)$ on any finite interval I of length at least $l_0 > 0$, then $|G'(Q)| \leq M$. Picking $\varphi(v) = \varphi_I(v) = K_I(1 + v^2)$ and using Equation (35) then yields the result. □

Propositions A.2 and A.3 establish that we can pick constants A, B, R_P, R_G and R_α and without loss of generality consider the system of ODEs

$$\hat{\Pi}''(Q) = \hat{F}_\Pi\left(Q, \hat{\Pi}(Q), \hat{\Pi}'(Q), \hat{G}'(Q)\right) \text{ and } \hat{G}''(Q) = \hat{F}_G\left(Q, \hat{G}(Q), \hat{G}'(Q), \hat{\Pi}'(Q)\right) \tag{36}$$

on finite intervals of $\mathbb{R}$ where

$$\hat{F}_\Pi(Q, u, v, x) = F_\Pi(Q, \min\{A, \max\{B, u\}\}, \min\{-R_P, \max\{v, 1/(2\alpha) - R_\alpha\}\}, \min\{-R_G, \max\{x, R_G\}\}$$

and

$$\hat{F}_G(Q, u, v, x) = F_\Pi(Q, \min\{A, \max\{B, u\}\}, \min\{-R_G, \max\{v, R_G\}\}, \min\{-R_P, \max\{x, 1/(2\alpha) - R_\alpha\}\}$$

Proposition A.4. *For any finite interval $[\underline{Q}, \bar{Q}] \subset \mathbb{R}$, the BVP (36) with boundary conditions*

$$\hat{\Pi}(\underline{Q}) = \underline{\Pi},\ \hat{\Pi}(\bar{Q}) = \bar{\Pi},\ \hat{G}(\underline{Q}) = \underline{G},\ \text{and } \hat{G}(\bar{Q}) = \bar{G}$$

[38] This is because we proved that for all Q, $\Pi'(Q) > -R$ for some $R > 0$.

has a solution.

Proof. The proof is an adaptation of the proof of Hartman (2002), Th. 4.2, p. 424. The proof relies on constructing an operator on the space of continuously differentiable functions on finite intervals and applying Schauder's Theorem (e.g. Hartman (2002), Th. 0.2, p. 405).

We can write the ODE system in vector form as

$$X''(Q) = \hat{F}(Q, X(Q), X'(Q))$$

where $X(Q) = (\Pi(Q), G(Q))$ and $\hat{F} = \left(\hat{F}_\Pi, \hat{F}_G\right)$. We now establish that this system has a solution on any finite interval $I \equiv [\underline{Q}, \bar{Q}] \subset \mathbb{R}$ with the above boundary conditions.

Let $l_I = \bar{Q} - \underline{Q}$ and denote with $\mathcal{D}_I$ the Banach space of continuously differentiable vector-valued functions $d(Q) : I \to \mathbb{R}^2$. Endow $\mathcal{D}_I$ with the norm

$$|d| = \max\left(\max_{Q\in I} \|d(Q)\|, \frac{l_I}{4} \max_{Q\in I} \|d'(Q)\|\right)$$

where $\|.\|$ is the Euclidean norm. Consider the subset $\mathcal{D}_I^R = \{d(Q) : \max_{Q\in I} |d(Q)| \leq R\}$ for some $R > 0$. Hartman (2002), Th. 3.1, p. 419 implies that the ODE system

$$X''(Q) = \hat{F}(Q, d(Q), d'(Q))$$

with boundary conditions $X(\underline{Q}) = \underline{X}$ and $X(\bar{Q}) = \bar{X}$, where $\underline{X} = (\underline{\Pi}, \underline{G})$ and $\bar{X} = (\bar{\Pi}, \bar{G})$, has a unique solution for any $d \in \mathcal{D}_I^R$, which is given by

$$X(Q) = -\int_{\underline{Q}}^{\bar{Q}} G(Q,s) \hat{F}(s, d(s), d'(s))\, ds + \left(Q - \underline{Q}\right) \frac{1}{l_I} \left(\bar{X} - \underline{X}\right) + \underline{X} \tag{37}$$

where

$$G(Q,s) = \frac{1}{l_I} \left(\left(\bar{Q} - Q\right)\left(s - \underline{Q}\right) \mathbf{1}_{\{s\leq Q\}} + \left(\bar{Q} - s\right)\left(Q - \underline{Q}\right) \mathbf{1}_{\{s>Q\}}\right)$$

Now, define the operator $T : \mathcal{D}_I \to \mathcal{D}_I$ such that $T(d(Q)) = X(Q)$. To apply Schauder's fixed point theorem, we must establish that T is a continuous and compact operator which maps $\mathcal{D}_I^R$ onto itself.

To establish that T maps $\mathcal{D}_I^R$ onto itself, note that it holds that

$$\int_{\underline{Q}}^{\bar{Q}} G(Q,s)\, ds \leq \frac{l_I^2}{8} \text{ and } \int_{\underline{Q}}^{\bar{Q}} \frac{\partial G(Q,s)}{\partial Q} \leq \frac{l_I}{2}.$$

Therefore,

$$\|X(Q)\| \leq \frac{l_I^2}{8} \max_{Q\in[\underline{Q},\bar{Q}]} \left\|\hat{F}(Q, d(Q), d'(Q))\right\| + \left\|\bar{X} - \underline{X}\right\|$$

and, by differentiating Equation (37) with respect to Q,

$$\|X'(Q)\| \leq \frac{l_I}{2} \max_{Q\in[\underline{Q},\bar{Q}]} \left\|\hat{F}(Q, d(Q), d'(Q))\right\| + \frac{1}{l_I} \left\|\bar{X} - \underline{X}\right\|.$$

By construction of $\hat{F}$, it holds that

$$\max_{Q\in[\underline{Q},\bar{Q}]} \left\|\hat{F}\left(Q, d\left(Q\right), d'\left(Q\right)\right)\right\| \leq K_I$$

and therefore

$$\left\|X\left(Q\right)\right\| \leq \frac{l_I^2}{8}K_I + \left\|\bar{X}-\underline{X}\right\| \text{ and } \left\|X'\left(Q\right)\right\| \leq \frac{l_I}{2}K_I + \left\|\bar{X}-\underline{X}\right\|$$

so that $|X\left(Q\right)| \leq \frac{l_I^2}{8}K_I + \left\|\bar{X}-\underline{X}\right\|$. Picking $R > \frac{l_I^2}{8}K_I + \left\|\bar{X}-\underline{X}\right\|$ ensures that T maps $\mathcal{D}_I^R$ into itself.

To establish continuity, pick $d_1, d_2 \in \mathcal{D}_I^R$ and let $X_1 = T\left(d_1\right)$ and $X_2 = T\left(d_2\right)$. Then,

$$|X_1 - X_2| \leq \frac{l_I}{4}\int_{\underline{Q}}^{\bar{Q}} \left\|\hat{F}\left(Q, d_1\left(Q\right), d_1'\left(Q\right)\right) - \hat{F}\left(Q, d_2\left(Q\right), d_2'\left(Q\right)\right)\right\| dQ,$$

so that $|d_1 - d_2| \to 0$ implies $|X_1 - X_2| \to 0$. Thus, T is continuous.

We finally establish that T is a compact operator, i.e. the range of T has a compact closure. Equation (37) implies that

$$\left\|X\left(Q_1\right) - X\left(Q_2\right)\right\| \quad \leq \quad K_I l_I \left|Q_1 - Q_2\right|$$

and

$$\left\|X'\left(Q_1\right) - X'\left(Q_2\right)\right\| \leq 2K_I \left|Q_1 - Q_2\right|,$$

which follows after some algebra. Hence, functions in the range of T are equicontinuous and the Arzela-Ascoli theorem implies that the range of T has a compact closure.

We can hence apply Schauder's theorem to T, which concludes the proof. □

It remains to extend the existence result in Proposition A.4 to the entire real line. To this end, consider a sequence of intervals $I_n = [\underline{Q}_n, \bar{Q}_n]$ for $n \in \mathbb{N}$ with $I_n \subset I_{n+1}$ and $\underline{Q}_n \to -\infty$ and $\bar{Q}_n \to \infty$ as $n \to \infty$. Consider the sequence of BVPs (36) on I_n with boundary conditions $\hat{\Pi}(\underline{Q}) = \underline{\Pi}_n$, $\hat{\Pi}(\bar{Q}) = \bar{\Pi}_n$, $\hat{G}(\underline{Q}) = \underline{G}_n$, and $\hat{G}(\bar{Q}) = \bar{G}_n$, such that as $n \to \infty$,

$$\underline{\Pi}_n \to \Pi\left(-\infty\right) = -\frac{\sigma^2}{r^2}\frac{\phi}{1-\theta}\left(S+\theta\right) \text{ and } \bar{\Pi}_n \to \Pi\left(\infty\right) = -\frac{\sigma^2}{r^2}\frac{\phi}{1-\theta}\left(S-\theta\right),$$

and

$$\underline{G}_n \to G\left(-\infty\right) = \frac{1}{2}\frac{\phi^2\sigma^2}{r}\left(\frac{1}{1-\theta}\left(S+\theta\right)\right)^2 \text{ and } \bar{G}_n \to G\left(\infty\right) = \frac{1}{2}\frac{\phi^2\sigma^2}{r}\left(\frac{1}{1-\theta}\left(S-\theta\right)\right)^2.$$

On any interval I_n, Proposition A.4 implies that there exists a solution $\left(\Pi_n\left(Q\right), G_n\left(Q\right)\right)$. We will use the Arzela-Ascoli theorem to construct a solution on $\mathbb{R}$ as a limit of $\left(\Pi_n\left(Q\right), G_n\left(Q\right)\right)$. To this end, we use Hartman (2002), Corollary 5.1, p. 431, which we reproduce below.

Corollary 3. *[Hartman (2002), Cor. 5.1] Suppose that a (vector-valued) function $X\left(Q\right)$ is* *twice continuously differentiable on the interval* $[\underline{Q},\bar{Q}]$ *with* $[\underline{Q},\tilde{Q}]\subset[\underline{Q},\bar{Q}]$ *for some* $\tilde{Q}$ *and*

$$\|X\|\leq R \text{ and } \|X''\|\leq K_1+K_2\|X'\|^2 \tag{38}$$

for some strictly positive constants R, K_1, *and* K_2. *Then, there exists a constant* M *such that* $\|X'\|\leq M$ *for all* $Q\in p=[\underline{Q},\bar{Q}]$. *The constant* M *depends only on* R, K_1, K_2, *and* $\tilde{Q}$.

From our construction of $\hat{F}$ in Equation (36), it follows that the conditions in Equation (38) hold on any finite interval $[\underline{Q},\bar{Q}]$. Fix the interval $[\underline{Q}_1,\bar{Q}_1]$. Propositions A.2 and A.3 imply that any solution $(\Pi_n(Q),G_n(Q))$ satisfies $\Pi_n(Q),G_n(Q)\in[A,B]$ for $Q\in[\underline{Q}_1,\bar{Q}_1]$, and Corollary 3 implies that $|\Pi_n'(Q)|,|G_n'(Q)|\leq M$ for $Q\in[\underline{Q}_1,\bar{Q}_1]$. Notably, M depends on $[\underline{Q}_1,\bar{Q}_1]$ but not on n. Then, given $\Pi_n(Q),G_n(Q),\Pi_n'(Q)$, and $G_n'(Q)$ are bounded on $[\underline{Q}_1,\bar{Q}_1]$, Equation (36) implies that $\Pi_n''(Q)$ and $G_n''(Q)$ are bounded on $[\underline{Q}_1,\bar{Q}_1]$ as well. Since the bounds do not depend on n, the sequence of functions $(\Pi_n(Q),G_n(Q))$ is bounded (uniformly in n) and equicontinuous. Hence, the Arzela-Ascoli theorem implies that there is a subsequence which converges uniformly to a continuously differentiable function $\left(\tilde{\Pi}_1,\tilde{G}_1\right)$ on $[\underline{Q}_1,\bar{Q}_1]$. Since the second derivatives are also bounded on $[\underline{Q}_1,\bar{Q}_1]$, uniformly in n, $(\Pi_n'(Q),G_n'(Q))$ are also equicontinuous, so that $\left(\tilde{\Pi},\tilde{G}\right)$ is twice continuously differentiable without loss of generality and satisfies Equation (36).

Now, pick $[\underline{Q}_2,\bar{Q}_2]$. Repeating the argument above, the original subsequence has another subsequence that converges uniformly to a limit $\left(\tilde{\Pi}_2,\tilde{G}_2\right)$ on $[\underline{Q}_2,\bar{Q}_2]$, and the limit function satisfies Equation (36) on $[\underline{Q}_2,\bar{Q}_2]$. Proceeding iteratively, we can cover the entire real line $\mathbb{R}$, which concludes our proof.

A.2 Proof of Proposition 2

Given the price conjecture in Equation (41), the Q-value and risk premium evolve according to Equations (15) and (14). The HJB equation of trader type i is given by

$$\begin{aligned}\delta V^i &= \max_{C_i,N_i} -\frac{1}{\phi}e^{-\phi C_i}+V_W^i\left(rW-C+N_i\left(\mu_P(Q)-rP(D,Q)+D\right)\right)\\ &\quad+\frac{1}{2}V_{WW}^i\left(N_i\sigma_P(Q)+Y_i\right)^2+V_Q^i\mu_Q(Q)+V_{QQ}\frac{1}{2}\sigma_Q^2(Q)+V_{WQ}^i\left(N_i\sigma_P(Q)+Y_i\right)\sigma_Q(Q).\end{aligned}$$

Using the conjecture $V^i(W,Q)=-\frac{1}{\phi r}\exp\left(-\phi rW-G_i(Q)-\frac{\delta-r}{r}\right)$, the FOC for N_i is given by

$$N_i(Q)=\frac{1}{\phi r\sigma_P^2(Q)}\left(\mu_P(Q)-rP(D,Q)+D\right)-\frac{2\alpha}{\phi r}G_i'(Q)-\frac{Y_i}{\sigma_P(Q)}.$$

Plugging the FOC for N_i and the FOC for C into the HJB equation, using $Y_i^2=Y^2$, and simplifying yields

$$\begin{aligned}rG_i(Q) &= \frac{1}{2}(\phi r)^2N_i^2(Q)\sigma_P^2(Q)-\frac{1}{2}(\phi rY)^2\\ &\quad+G_i'(Q)\mu_Q(Q)+\left(G_i''(Q)-G_i'(Q)^2\right)\frac{1}{2}\alpha^2\sigma_P^2(Q)-G_i'(Q)\phi r\sigma_Q(Q)Y_i.\end{aligned}$$

The market clearing condition is given by

$$mN^S(Q) + (1-m)N^L(Q) = \frac{1}{1-\theta}\left(S - \theta N^Q(Q)\right).$$

Plugging in the optimal demands and rearranging then yields Equation (13). We can rewrite that equation as

$$\begin{aligned} \mu_P(Q) - rP(D,Q) + D &= \Pi'(Q)\mu_Q(Q) + \frac{1}{2}\Pi''(Q)\sigma_Q^2(Q) - r\Pi(Q) \\ &= \frac{\phi r \sigma_P^2(Q)}{1-\theta}\left(S - \theta N^Q(Q)\right) + \bar{G}'(Q)\sigma_P(Q)\sigma_Q(Q) + \phi r \bar{Y}\sigma_P(Q) \end{aligned}$$

and then substitute into $N_i(Q)$, which yields Equation (18). Plugging that equation into Equation (40) then yields Equation (12).

Inspecting Equations (12) and (13), the only difference to Equations (24) and (25) are the terms $G_i'(Q)\phi r\sigma_Q(Q)Y_i - \frac{1}{2}(\phi r Y)^2$ and $\bar{G}'(Q)\sigma_P(Q)\sigma_Q(Q) + \phi r\bar{Y}\sigma_P(Q)$, respectively. The existence proof of Proposition A.1 then carries through with only minor modifications. In particular, fixing two functions $(G_1(Q), G_2(Q))$ with uniformly bounded derivatives implies that $\bar{G}'(Q)$ is uniformly bounded, so that we can apply Proposition A.2 to Equation (13) replacing $G'(Q)$ with $\bar{G}'(Q)$. The argument in Lemma 1 applies with minor modifications, which establishes that $\sigma_P(Q) \geq \underline{\sigma} > 0$ for all Q and that $\sigma_P(Q)$ is uniformly bounded. Then, we can apply the argument in Proposition A.3 separately to the functions $G_1(Q)$ and $G_2(Q)$. Thus, Propositions A.2 and A.3 imply that we can a priori assume bounded and continuous solutions to the ODEs (12) and (13). This is the main part of the existence argument in Proposition A.4. The remainder of the proof of Proposition A.4 holds for an arbitrary dimensional ODE system, and thus continues to apply.

A.3 Proof of Proposition 3

We break the argument down into multiple steps.

1. *Inferring $\Pi(Q_t)$ and $\alpha\Pi'(Q_t)$ from dividends and prices.* Traders observe the dividend process D_t, and, therefore, the path of the brownian motion B_t, since $D_t = \mu t + \sigma B_t$. Since traders also observe the price path, they infer the price markup $\Pi(Q_t)$ (see Equation (5)). From observing the price path, traders also infer the price volatility $\sigma_P(Q_t)$,[39] which implies that they infer $\alpha\Pi'(Q_t)$.

2. *Inferring Q-learner demand $N^Q(Q_t)$, Q_t/β, and the ratio α/β from asset demands.* Since asset supply S is fixed and traders know their own demand N_t^R, they know $N^Q(Q_t)$ at each time t, i.e. they know the path of the Q-learner demand up to any time t. Equation (11) implies that this is equivalent to knowing Q_t/β for all t. From knowing the path of $N^Q(Q_t)$, they similarly infer the volatility of $N^Q(Q_t)$, which is given by
$$\frac{\alpha}{2\beta}\left(1 + N^Q(Q_t)\right)\left(1 - N^Q(Q_t)\right)\sigma_P(Q_t).$$

[39]Intuitively, observing path of P_t is equivalent to observing the path of P_t^2, and Ito's Lemma implies that $dP_t^2 - 2P_t dP_t = \sigma_P^2(Q_t)\,dt$, so traders can infer $\sigma_P(Q_t)$.

Since both $N^Q(Q_t)$ and $\sigma_P(Q_t)$ are known, traders infer the ratio α/β.

3. *Inferring $\alpha G'(Q_t)$ and $\alpha^2\Pi''(Q_t)$ from realized returns.* Knowing both $\sigma_P(Q_t)$ and the path of B_t, traders can infer $\alpha G'(Q_t)$ from observing the path of realized returns R_t. Since traders know the price volatility, they can also infer excess volatility. Excess volatility itself follows a diffusion process, which is given by

$$d\sigma_P(Q_t) = \begin{array}{r} \left(\frac{2\alpha\sigma\Pi''(Q_t)}{r(1-2\alpha\Pi'(Q_t))^2}\mu_Q(Q) + \frac{1}{2}\frac{2\alpha\sigma\left(4\alpha\Pi''(Q_t)^2+\Pi'''(Q_t)(1-2\alpha\Pi'(Q_t))\right)}{r(1-2\alpha\Pi'(Q_t))^3}\sigma_Q^2(Q_t)\right) dt \\ +\frac{2\alpha\sigma\Pi''(Q_t)}{r(1-2\alpha\Pi'(Q_t))^2}\sigma_Q(Q_t)\,dB_t. \end{array}$$

From this process, traders can infer the "volatility of excess volatility,"

$$\alpha\sigma'_{\Delta P}(Q_t)\,\sigma_P(Q_t) = \frac{\sigma}{r}\frac{4\alpha^2\Pi''(Q_t)}{(1-2\alpha\Pi'(Q_t))^2}\sigma_P(Q_t).$$

Hence, traders can infer $\alpha^2\Pi''(Q_t)$.

4. *Inferring αQ_t from the first-order constraint.* Traders know their own demand N_t^R and understand that it must satisfy the first-order constraint (22), which implies that they know

$$\frac{\mu_P(Q_t) - \frac{\mu}{r} - r\Pi(Q_t) - \alpha G'(Q_t)\,\sigma_P^2(Q_t)}{\phi r\sigma_P^2(Q_t)}.$$

Using Equation (17), and their knowledge of $\alpha\Pi'(Q_t)$, $\alpha^2\Pi''(Q_t)$, and $\alpha G'(Q_t)$, traders can then back out the value of αQ_t.

5. *Inferring (α, β, Q_0).* We have now established that traders know α/β and the paths of Q_t/β and αQ_t. Hence, they can infer α, β, and Q_t for all t. Thus, the hyperparameters (α, β, Q_0) are common knowledge without loss of generality.

Online Appendix for "Trading against Algorithms: Price Dynamics and Risk Sharing in a Market with Q-learners"

B Convergence of discrete time Q-learner demand

In single-agent environments, the Q-learning algorithm is guaranteed to converge to the true Q-matrix $Q(s,a)$ under mild conditions, which among others guarantee that each action is sampled infinitely many times in each state over an infinite horizon. However, in more general settings, such convergence is not guaranteed.

In this Appendix, we show that the Q-learner demand from the discrete time specification converges to the continuous time specification that we assume in Section 3.1. The discrete-time analog of Equation (1) is given by

$$D_{t+1} - D_t = \mu + \sigma\varepsilon_{t+1},$$

where $\varepsilon_t \sim N(0,1)$ is iid.

In what follows, we conjecture that the equilibrium price can be expressed as

$$P_t = \frac{1}{r}\left(D_t + \frac{\mu}{r}\right) + \Pi(Q_t), \tag{41}$$

where $\Pi(Q)$ is a twice continuously differentiable function. Then, a Q-learner's realized return is given by

$$R_{t+1} = P_{t+1} - P_t + D_t - rP_t = \frac{\sigma}{r}\varepsilon_{t+1} + \Pi(Q_{t+1}) - (1+r)\Pi(Q_t)$$

and the evolution of the Q-value is given by

$$Q_{t+1} - Q_t \;\;=\;\; 2\alpha\left(\frac{\sigma}{r}\varepsilon_{t+1} + \Pi(Q_{t+1}) - (1+r)\Pi(Q_t) - \frac{1}{2}Q_t\right).$$

Note that the evolution of Q-values exhibits a dynamic feedback effect. Specifically, next period's Q-value Q_{t+1} is determined by future price markup $\Pi(Q_{t+1})$, as the price markup affects the realized return. However, the future markup is determined by the future Q-value Q_{t+1},and so on. As a result, the continuous-time limit is well defined only if $\Pi'(Q) < \frac{1}{2\alpha}$ for all Q. To see why heuristically, take a small change in Q_t, which implies that

$$\frac{Q_{t+1} - 2\alpha\Pi(Q_{t+1}) - (Q_t - 2\alpha\Pi(Q_t))}{Q_{t+1} - Q_t} \approx 1 - 2\alpha\Pi'(Q_t),$$

and so we can express Equation (42) as

$$Q_{t+1} - Q_t \approx \frac{2\alpha}{1 - 2\alpha\Pi'(Q_t)}\left(\frac{\sigma}{r}\varepsilon_{t+1} - r\Pi(Q_t) - \frac{1}{2}Q_t\right).$$

This implies that as $\Pi'(Q_t) \to \frac{1}{2\alpha}$, the RHS explodes because the noise in the dividend process is amplified infinitely.

As such, in the following result, we impose $\Pi'(Q) < \frac{1}{2\alpha}$ as a condition. However, we verify this condition holds in equilibrium in Proposition A.1.

Proposition B.1. *Suppose that* $\Pi(Q)$ *is twice continuously differentiable and that* $\sup_{Q\in\mathbb{R}} \Pi'(Q) < \frac{1}{2\alpha}$. *Then, as* $h \to 0$, *the approximation*

$$Q_h^*(t) = \sqrt{h}\left(Q_{\lfloor t/h \rfloor} + \left(t - h\left\lfloor \frac{t}{h} \right\rfloor\right)\left(Q_{\lfloor t/h \rfloor+1} - Q_{\lfloor t/h \rfloor}\right)\right) \tag{43}$$

converges in distribution to the unique weak solution of the stochastic differential equation (SDE)

$$Q_t = Q_0 + \int_0^t \mu_Q(Q_s)\,ds + \int_0^t \sigma_Q(Q_s)\,dB_s,$$

where

$$\mu_Q(Q_t) = \frac{2\alpha}{1 - 2\alpha\Pi'(Q_t)}\left(\frac{1}{2}\sigma_Q(Q_t)^2\,\Pi''(Q_t) - r\Pi(Q_t) - \frac{1}{2}Q_t\right)$$

and

$$\sigma_Q(Q_t) = \frac{\sigma}{r}\frac{2\alpha}{1 - 2\alpha\Pi'(Q_t)}.$$

Similarly, the approximation of the price P_t converges to Equation (14).

Corollary 4. *Under Conjecture (41), as* $h \to 0$ *the approximation*

$$P_h^*(t) = \sqrt{h}\left(P_{\lfloor t/h \rfloor} + \left(t - h\left\lfloor \frac{t}{h} \right\rfloor\right)\left(P_{\lfloor t/h \rfloor+1} - P_{\lfloor t/h \rfloor}\right)\right)$$

converges in distribution to the unique weak solution of the SDE

$$P_t = \frac{\mu}{r^2} + \int_0^t \mu_P(Q_s)\,ds + \int_0^t \sigma_P(Q_s)\,dB(s)$$

where

$$\mu_P(Q_t) = \frac{\mu}{r} + \Pi'(Q_t)\,\mu_Q(Q_t) + \frac{1}{2}\Pi''(Q_t)\,\sigma_Q^2(Q_t)$$

and

$$\sigma_P(Q_t) = \frac{\sigma}{r}\frac{1}{1 - 2\alpha\Pi'(Q_t)}.$$

In particular, for all $t \geq 0$,

$$P_t = \frac{\mu}{r^2} + \frac{1}{r}D_t + \Pi(Q_t).$$

Proof. This follows from applying Ito's Lemma to Conjecture (41). □

B.1 Proof of Proposition B.1

Rewrite Equation (42) as

$$Q_{t+1} - 2\alpha\Pi(Q_{t+1}) = Q_t - 2\alpha\Pi(Q_t) + \frac{\alpha\sigma}{r}\varepsilon_{t+1} + \mu(Q_t),$$

where $\mu(Q) = -\alpha Q - 2\alpha r \Pi(Q)$. Define

$$\tilde{Q}_t = H(Q_t) = Q_t - \alpha \Pi(Q_t).$$

Given the assumption $\sup_{Q\in\mathbb{R}} \Pi'(Q) < 1/(2\alpha)$, $H(Q)$ is invertible. Then, we can write

$$\tilde{Q}_{t+1} = \tilde{Q}_t + \frac{2\alpha\sigma}{r}\varepsilon_{t+1} + \tilde{\mu}\left(\tilde{Q}_t\right)$$

where $\tilde{\mu}\left(\tilde{Q}_t\right) = \mu\left(H^{-1}\left(\tilde{Q}_t\right)\right)$. Define a grid with step size h between any points t and t'. Indexing points on the grid with k, define the approximation

$$\tilde{Q}_{k+1,h} = \tilde{Q}_{k,h} + h\tilde{\mu}\left(\tilde{Q}_{k,h}\right) + \sqrt{h}\frac{2\alpha\sigma}{r}\varepsilon_{k+1,h},$$

where $\varepsilon_{k,h} \sim N(0,1)$ for all k and h. Finally, define the linear interpolation as $\tilde{Q}_h^*(t)$, where

$$\tilde{Q}_h^*(t) = \sqrt{h}\left(\tilde{Q}_{\lfloor t/h\rfloor,h} + \left(t - h\left\lfloor\frac{t}{h}\right\rfloor\right)\left(\tilde{Q}_{\lfloor t/h\rfloor+1,h} - \tilde{Q}_{\lfloor t/h\rfloor,h}\right)\right).$$

The function $\tilde{Q}_h^*(t)$ defines a Markov process which is deterministic at all points $t \in (kh,(k+1)h)$ and for which the transition probabilities times $t \in \{kh\}_{i\in\mathbb{N}}$ are defined by

$$\Pr\left(\tilde{Q}_h^*((k+1)h) \in \Gamma | \tilde{Q}_h^*(kh) = x\right) = \Pi_h(x,\Gamma)$$

for any $\Gamma \in \mathcal{B}(\mathbb{R})$. Since $\varepsilon_{k,h}$ is normally distributed, $\tilde{Q}_h^*(t)$ admits a transition density, which is given by

$$\Pi_h(x,y) = \frac{1}{\sqrt{2\pi h}\tilde{\sigma}}\exp\left(-\frac{1}{2}\frac{(y - x - h\tilde{\mu}(x))^2}{h\tilde{\sigma}^2}\right),$$

where we write $\tilde{\sigma} = 2\alpha\sigma/r$ to save notation.

Let $C_0^\infty(\mathbb{R})$ denote the space of infinitely continuously differentiable functions $f : \mathbb{R} \to \mathbb{R}$ with compact support. For any $f \in C_0^\infty(\mathbb{R})$,define

$$A_h f = \frac{1}{h}\int (f(y) - f(x))\,\Pi_h(x,y)\,dy.$$

Lemma 2. *For any* $f \in C_0^\infty(\mathbb{R})$

$$A_h f \to \mathcal{L}f$$

uniformly on compact subsets of $\mathbb{R}$ *as* $h \to 0$*, where*

$$\mathcal{L}f(x) = \tilde{\mu}(x) f'(x) + \frac{1}{2}\tilde{\sigma}^2 f''(x).$$

Proof. Since $\mathbb{R}$ is locally compact, to establish uniform convergence on compact subsets, it is sufficient to establish locally uniform convergence, i.e. for all $x \in \mathbb{R}$, there exists an $\varepsilon > 0$

such that

$$\lim_{h\to 0}\sup_{x'\in B(x,\varepsilon)}\left|A_h f\left(x'\right)-\mathcal{L}f\left(x'\right)\right|=0$$

for any $f\in C_0^\infty(\mathbb{R})$, where $B\left(x,\varepsilon\right)$ is the ε-ball around x.

Since we assumed that $\Pi\left(.\right)$ is continuously differentiable, $\tilde{\mu}\left(.\right)$ is locally Lipschitz continuous, i.e. there exists a $K>0$ such that $\left|\tilde{\mu}\left(x'\right)-\tilde{\mu}\left(x\right)\right|\leq K\left|x'-x\right|$ for all $x'\in B\left(x,\varepsilon\right)$, and in particular, $\left|\tilde{\mu}\left(x'\right)-\tilde{\mu}\left(x\right)\right|\leq K\varepsilon$.

Since f is infinitely differentiable, Taylor's theorem implies that

$$f\left(y\right)=\sum_{l=0}^{\infty}\frac{1}{l!}f^{(l)}\left(x\right)\left(y-x\right)^l.$$

Thus,

$$A_h f=\frac{1}{h}\left(f'\left(x\right)E_h\left[y-x\right]+\frac{1}{2}f''\left(x\right)E_h\left[\left(y-x\right)^2\right]+\sum_{l=3}^{\infty}\frac{1}{l!}f^{(l)}\left(x\right)E_h\left[\left(y-x\right)^l\right]\right)$$

where $E_h\left[.\right]$ is the expectations operator under Π_h. We have $E_h\left[y-x\right]=h\tilde{\mu}\left(x\right)$ for all x, which in particular implies that

$$\lim_{h\to 0}\sup_{x'\in B(x,\varepsilon)}\left|\frac{1}{h}E_h\left[y-x'\right]-\tilde{\mu}\left(x'\right)\right|=0.$$

Similarly,

$$\begin{aligned}
E_h\left[\left(y-x\right)^2\right] &= E_h\left[\left(y-x-h\tilde{\mu}\left(x\right)+h\tilde{\mu}\left(x\right)\right)^2\right]\\
&= E_h\left[\left(y-x-h\tilde{\mu}\left(x\right)\right)^2\right]+2E_h\left[\left(y-x-h\tilde{\mu}\left(x\right)\right)\right]h\tilde{\mu}\left(x\right)\\
&\quad +h^2\tilde{\mu}\left(x\right)^2
\end{aligned}$$

and

$$\begin{aligned}
\lim_{h\to 0}\sup_{x'\in B(x,\varepsilon)}\left|\frac{1}{h}E_h\left[y-x'\right]^2-\tilde{\sigma}\right| &= \lim_{h\to 0}\sup_{x'\in B(x,\varepsilon)}\left|h^2\tilde{\mu}^2\left(x'\right)\right|\\
&\leq \lim_{h\to 0}h^2K^2\varepsilon^2\\
&= 0.
\end{aligned}$$

Proceeding similarly for the higher order terms, we have

$$\begin{aligned}
E_h\left[\left(y-x\right)^3\right] &= E_h\left[\left(y-x-h\tilde{\mu}\left(x\right)+h\tilde{\mu}\left(x\right)\right)^3\right]\\
&= E_h\left[\left(y-x-h\tilde{\mu}\left(x\right)\right)^3\right]+3h^2\tilde{\mu}\left(x\right)\tilde{\sigma}^2\\
&= +3h^2\tilde{\mu}\left(x\right)^2E_h\left[\left(y-x-h\tilde{\mu}\left(x\right)\right)\right]+h^3\tilde{\mu}\left(x\right)^3.
\end{aligned}$$

Stein's Lemma[40] implies that

$$E_h\left[(y-x-h\tilde{\mu}(x))^3\right] = 2h\tilde{\sigma}^2 E_h\left[y-x-h\tilde{\mu}(x)\right] = 0$$

and thus

$$E_h\left[(y-x)^3\right] = 3h^2\tilde{\mu}(x)\,\tilde{\sigma}^2 + h^3\tilde{\mu}(x)^3$$

which implies that

$$\begin{aligned}\lim_{h\to 0}\sup_{x'\in B(x,\varepsilon)}\left|\frac{1}{h}E_h\left[(y-x)^3\right]\right| &\leq \lim_{h\to 0} 3h^2\tilde{\sigma}^2K\varepsilon + h^3K^3\varepsilon^3 \\ &= 0\end{aligned}$$

as $h\to 0$. Similarly,

$$\begin{aligned}E_h\left[(y-x)^4\right] &= E_h\left[(y-x-h\tilde{\mu}(x)+h\tilde{\mu}(x))^4\right] \\ &= E_h\left[(y-x-h\tilde{\mu}(x))^4\right] + 4h\tilde{\mu}(x)\,E_h\left[(y-x-h\tilde{\mu}(x))^3\right] \\ &\quad +6h^4\tilde{\mu}(x)^2\,\tilde{\sigma}^2 + 4h^3\tilde{\mu}(x)^3\,E_h\left[(y-x-h\tilde{\mu}(x))\right] \\ &\quad +h^4\tilde{\mu}(x)^4\,.\end{aligned}$$

Using Stein's Lemma again implies that

$$\begin{aligned}E_h\left[(y-x-h\tilde{\mu}(x))^4\right] &= 3h\tilde{\sigma}^2E_h\left[(y-x-h\tilde{\mu}(x))^2\right] \\ &= 3h^2\tilde{\sigma}^4,\end{aligned}$$

so that

$$\begin{aligned}\lim_{h\to 0}\sup_{x'\in B(x,\varepsilon)}\left|\frac{1}{h}E_h\left[(y-x)^4\right]\right| &\leq \lim_{h\to 0} 3h^2\tilde{\sigma}^4 + 6h^4K^2\varepsilon^2\tilde{\sigma}^2 + h^4K^4\varepsilon^4 \\ &= 0.\end{aligned}$$

Proceeding inductively, it follows that

$$\lim_{h\to 0}\sup_{x'\in B(x,\varepsilon)}\frac{1}{h}\sum_{l=3}^{\infty}\frac{1}{l!}f^{(l)}(x)\,E_h\left[(y-x)^l\right] = 0$$

as $h\to 0$ and therefore

$$A_hf(x)\to f'(x)\,\tilde{\mu}(x) + \frac{1}{2}f''(x)\,\tilde{\sigma}^2 = \mathcal{L}f$$

uniformly on compact subsets. □

We now apply Stroock and Varadhan (1997), Th. 11.2.3. Define with P_h the probability measure induced by $\tilde{Q}_h^*(t)$ on $C([0,\infty),\mathbb{R})$, the space of continuous trajectories from $[0,\infty)$ into $\mathbb{R}$. Similarly, define with P the probability measure on $C([0,\infty),\mathbb{R})$ which solves the

[40] For any differentiable function $g(X)$ and $X\sim N\left(\mu,\sigma^2\right)$, $E\left[g(X)(X-\mu)\right] = \sigma^2E\left[g'(X)\right]$.

martingale problem for $\mathcal{L}$.[41] The theorem states that if the martingale problem for $\mathcal{L}$ has a unique solution, and for all $f \in C_0^\infty(\mathbb{R})$, $A_h f \to \mathcal{L}f$ uniformly on compact subsets of $\mathbb{R}$, then $P_h \to P$ uniformly on compact subsets of $\mathbb{R}$.

The previous lemma establishes the necessary convergence. It only remains to show that the martingale problem for $\mathcal{L}$ has a unique solution. Kurtz (2010), Th. 1, states that the martingale problem for $\mathcal{L}$ has a unique solution if and only if the SDE associated with $\mathcal{L}$ has a unique weak solution. The SDE

$$\tilde{Q}_t = \int_0^t \tilde{\mu}\left(\tilde{Q}_s\right) ds + \tilde{\sigma} B_t \tag{44}$$

has a constant and positive volatility $\tilde{\sigma}$, and $\tilde{\mu}(\cdot)$ is continuous and therefore bounded on compact subsets of $\mathbb{R}$. Yong and Zhou (1999), Th. 6.1.2 then establishes that Equation (44) has a unique weak solution. Hence, Stroock and Varadhan (1997), Th. 11.2.3 implies that $P_h \to P$, or equivalently

$$\tilde{Q}_h^*(t) \to^d \tilde{Q}_t$$

on compact subsets of $\mathbb{R}$.

It remains to transform $\tilde{Q}_t$ back to Q_t. By the continuous mapping theorem, $Q_h^*(t)$, the linear interpolation of Q_t in Equation (43), converges in distribution to

$$Q_t = H^{-1}\left(\tilde{Q}_t\right).$$

Since $\tilde{Q}_t$ is a diffusion process, we can use Ito's Lemma and the implicit function rule to obtain

$$\begin{aligned}
dQ_t &= \left(H^{-1\prime}\left(\tilde{Q}_t\right)\tilde{\mu}\left(\tilde{Q}_t\right) + H^{-1\prime\prime}\left(\tilde{Q}_t\right)\frac{1}{2}\left(\frac{2\alpha\sigma}{r}\right)^2\right)dt + H^{-1\prime}\left(\tilde{Q}_t\right)2\alpha\frac{\sigma}{r}dB_t \\
&= \frac{2\alpha}{1-2\alpha\Pi'(Q_t)}\left(-r\Pi(Q_t) - \frac{1}{2}Q_t + \frac{\Pi''(Q_t)}{(1-2\alpha\Pi'(Q_t))^2}\frac{1}{2}\left(\frac{2\alpha\sigma}{r}\right)^2\right)dt \\
&\quad + \frac{1}{1-2\alpha\Pi'(Q_t)}\frac{2\alpha\sigma}{r}dB_t \\
&= \mu_Q(Q_t)\,dt + \sigma_Q(Q_t)\,dB_t.
\end{aligned}$$

Given our assumption $\sup_{Q\in\mathbb{R}} \Pi'(Q) < 1/(2\alpha)$, it holds that $\sigma_Q(Q) \geq \underline{\sigma}_Q > 0$ for all $Q \in \mathbb{R}$ and $|\mu_Q(Q)| < \infty$ for all $Q \in \mathbb{R}$, i.e. the SDE for Q_t is uniformly elliptic and has locally bounded drift. It is hence well-posed.

C Hyperparameter choice

In the baseline model, we assumed that the parameters of the Q-learning algorithm are exogenous. In reality, these parameters are determined by traders who choose to delegate their trading to an algorithm. In this section, we assume that there is a representative

[41] See Stroock and Varadhan (1997), p. 138 for a definition of martingale problems.

algorithmic trader with CARA utility, who delegates trading to the algorithm[42] and optimally chooses the learning rate α and the temperature parameter β. That is, the algorithmic trader determines how fast the algorithm learns and how aggressively it trades to maximize her utility, anticipating the resulting equilibrium. The proposition below shows that there exists a Q-REE in this setting. This equilibrium takes the same form as in Proposition A.1, except that α and β are now endogenous.

Specifically, the algorithmic trader's problem is now given by

$$V^D(W_0, Q_0) = \sup_{\{C_s\}_{s\geq 0}, \alpha\in[0,1], \beta\geq 0} -\mathbb{E}_t\left[\int_0^\infty e^{-\delta(s-t)}\frac{1}{\phi}e^{-\phi C_s}ds\right]$$

subject to

$$dW_t = \big(rW_t - C_t + N^Q(Q_t)(D_t - rP_t)\big)dt + N^Q(Q_t)\,dP_t$$

and

$$dQ_t = \alpha dR_t - \frac{\alpha}{2}Q_t dt.$$

That is, the trader chooses α and β at time zero and chooses how much to consume at each time, but lets the algorithm trade on his behalf. Here, note that $N^Q(Q)$ depends implicitly on β (see Equation (11)).[43] All other traders behave as in the baseline model.

Proposition C.1. *A Q-REE in which the large trader delegates trading to the algorithm and chooses α and β optimally exists and takes the same form as the equilibrium in Proposition A.1. The delegating trader's value function is given by*

$$V^D(W, Q) = -\frac{1}{\phi r}\exp\left(-\phi rW - G_D(Q) - \frac{\delta - r}{r}\right),$$

where $G_D(Q)$ is determined by the ODE

$$\begin{aligned} rG_D(Q) = {} & \phi r N^Q(Q)\left(\mu_P(Q) - \frac{\mu}{r} - rP(Q)\right) \\ & -\frac{1}{2}(\phi r)^2\sigma_P^2(Q)N^Q(Q)^2 \\ & +G_D'(Q)\big(\mu_Q(Q) - \alpha\phi r\sigma_P^2(Q)N^Q(Q)\big) \\ & +\frac{1}{2}\left(G_D''(Q) - G_D'(Q)^2\right)\alpha^2\sigma_P^2(Q) \end{aligned}$$

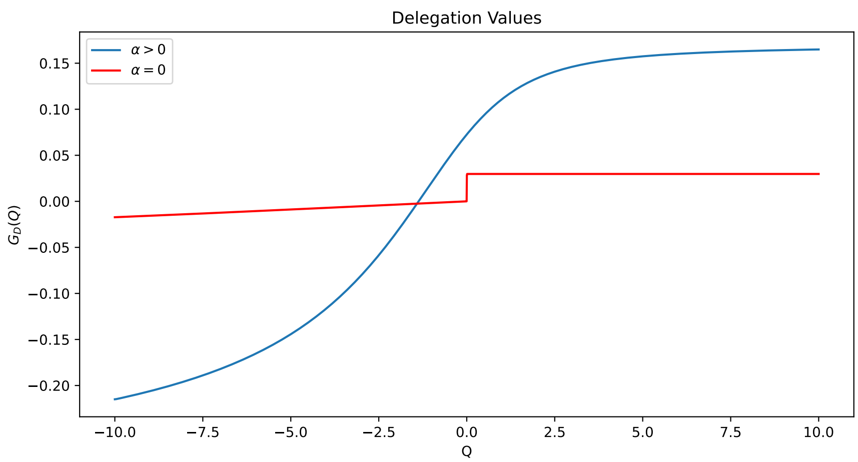
with boundary conditions $G_D'(-\infty) = G_D'(\infty) = 0$, and $P(Q)$ and $G(Q)$ satisfy Equations (24) and (25).

[42] This is optimal if trading "rationally" is costly. For example, hedge funds may need to hire quants and traders and share profits with them to incentivize them to gather information and trade. Trading via an algorithm may be cheaper, and it is optimal to delegate whenever $V^D(W_0 - I, Q_0) \geq V(W_0, Q_0)$, where V^D is the trader's expected value from delegating and I is the cost.

[43] As we have shown in Section 4.3, traders learn the hyperparameters from observing equilibrium prices over an arbitrarily small time horizon. Thus, without loss of generality, we assume that the trader's choice of α and β is public.

Figure 11: Delegation Values for $\alpha = 0$ and $\alpha > 0$

The figure plots the delegation values $G_D(Q)$ for the large trader for $\alpha = 0$ (red) and $\alpha > 0$ (blue). Whenever the blue line is above the red line, it is optimal to set $\alpha > 0$ for a given Q_0. Parameter values are set to: $r = 0.05$, $\phi = 1$, $\sigma = 0.075$, $\alpha = 0.1$ and $\beta = 1$ (for the blue line), $\theta = 0.3$, and $S = 5$.



Proof. The large trader's HJB equation is given by

$$\begin{aligned}\delta V^D &= \max_C -\frac{1}{\phi}e^{-\phi C} + V_W^D\left(rW - C + N^Q(Q)\left(\mu_P(Q) - \frac{\mu}{r} - rP(Q)\right)\right) \\ &\quad + \frac{1}{2}V_{WW}^D\sigma_P^2(Q)\,N^Q(Q)^2 + V_Q^D\mu_Q(Q) + \frac{1}{2}V_{QQ}^D\alpha^2\sigma_P^2(Q) \\ &\quad + V_{WQ}^D\alpha\sigma_P^2(Q)\,N^Q(Q).\end{aligned}$$

Conjecturing an equilibrium of the same form as in Proposition A.1, plugging the FOC for consumption $e^{-\phi C} = V_W^D(Q)$, and rearranging yields Equation (45). Existence of the equilibrium follows from the same argument as for Proposition A.1, which established existence for arbitrary values of α and β. □

Whenever $|S|$ is sufficiently large, the Q-learner makes positive profits. In Figure 11, we provide a numerical example when it is optimal to set $\alpha > 0$ and $\beta > 0$, so that in equilibrium, the Q-value indeed reacts to price changes. The kink in the red line occurs because for $S > 0$, buying is profitable, but when $Q_0 < 0$, $N^Q \leq 0$. It is then optimal to set $\beta = \infty$, so that $N^Q = 0$. By contrast, when $Q_0 > 0$, $N^Q \geq 0$ and it is optimal to set $\beta = 0$ which ensures that $N^Q = 1$, i.e. the policy is greedy and the algorithm buys as much as possible.

D Replacing Q-learners with a risk-neutral trader

We now consider a benchmark in which the Q-learners are replaced with a risk-neutral trader who chooses trades optimally and is a price taker.[44] All other assumptions in the model are unchanged, and, in particular, the risk-neutral trader is subject to choosing $N_t^N \in [-1, 1]$. This model variant features multiple equilibria and is prone to sunspots. For example, depending on S and θ, there may be an equilibrium where the risk-neutral trader always buys a share, but also another equilibrium where the risk-neutral trader sells a share.

Ignoring sunspots, however, any equilibrium is simple and resembles the equilibrium in Proposition 1. It features a constant risk premium Π, so that

$$P\left(D_t\right) = \frac{1}{r} D_t + \frac{\mu}{r^2} + \Pi.$$

Either $D_t + \mu/r - rP_t > 0$ and the risk-neutral trader buys one unit, i.e. $N_t^N = 1$; or $D_t + \mu/r - rP_t < 0$, and the risk-neutral trader sells one unit, i.e. $N_t^N = -1$; or $D_t + \mu/r - rP_t = 0$ and the risk-neutral trader is indifferent.

Proposition D.1. *If $S < \theta$, there exists an REE in which $\Pi < 0$ and $N_t^N = 1$ for all t. If $S > -\theta$, there exists an REE in which $\Pi > 0$ and $N_t^N = -1$ for all t. If $S \in [-\theta, \theta]$, there exists an REE in which $\Pi = 0$ for all t.*

Proof. Conjecture that $\Pi < 0$. Then, $N_t^N = 1$ and the market clearing condition becomes $(1-\theta) N_t^R + \theta = S$. Together with the FOC of the traders with CARA utility, this implies that

$$\Pi = \frac{\phi \sigma_{P0}^2}{1-\theta} \left(S - \theta\right).$$

Thus, $\Pi < 0$ whenever $S < \theta$, otherwise, this cannot be an equilibrium. Next, conjecture that $\Pi > 0$. Then, $N_t^N = -1$ and the market clearing condition is $(1-\theta) N_t^R - \theta = S$, which implies that

$$\Pi = \frac{\phi \sigma_{P0}^2}{1-\theta} \left(S + \theta\right).$$

Thus, $\Pi > 0$ whenever $S > -\theta$. Finally, conjecture that $\Pi = 0$. Then, it must be the case that

$$\Pi = \frac{\phi \sigma_{P0}^2}{1-\theta} \left(S - \theta N^N\right)$$

so that $N^N = S/\theta$. □

In particular for $|S| > \theta$, there exists a unique REE in which the risk premium is constant, in line with Proposition 1. For $|S| < \theta$, there may be sunspots. With Q-learners, any variation in risk premia is driven by changes in the Q-value, whereas here, any potential change in the risk premium is due to sunspots. Thus, the two models generate price changes for fundamentally different reasons.

[44] Equivalently, there exists a continuum of rational investors of mass θ.

E Biased traders

In Section 6, Q-learners realized profits by providing liquidity to human traders with hedging motives, and simultaneously improved human traders' welfare. In this section, we instead consider biased human traders. Q-learners then may learn to exploit human traders' biases, and realize profits at their expense.

As in Section 3, the total mass of human traders is $1-\theta$. A fraction $m \in [0,1]$ of traders is optimistic and mass $1-m$ is pessimistic. Let $\mathbb{P}$ denote the measure under which B_t is a brownian motion.[45] We take $\mathbb{P}$ to be the true measure under which the dividend process evolves, and we define optimism and pessimism via Girsanov's change of measure. Specifically, human traders of type $i \in \{O,P\}$ have a subjective probability measure $\mathbb{P}^i$ under which

$$B_t^i = B_t - \frac{\hat{\mu}_i}{\sigma} t$$

is a Brownian motion, where $\hat{\mu}_O > 0 > \hat{\mu}_P$. Thus, optimistic traders believe that the drift in the dividend process is larger than in reality and pessimistic traders believe it is smaller, i.e.

$$dD_t = (\mu + \hat{\mu}_i)\, dt + \sigma dB_t^i$$

under $\mathbb{P}^i$. Both types of traders evaluate the evolution of prices and the evolution of Q-values under their subjective belief $\mathbb{P}^i$. They are otherwise rational, i.e. they understand the law of motion for Q-values (15) and for prices (41), and they choose their demand and their consumption optimally given their belief.

The following result characterizes the equilibrium.

Proposition E.1. *There exists a Q-REE in which subjective gains from trade are given by*

$$\begin{aligned} rG_i(Q) &= \frac{1}{2}(\phi r)^2 \sigma_P^2(Q) N_i^2(Q) + G_i'(Q)\left(\mu_Q(Q) + \alpha \Pi'(Q)\sigma_Q(Q)\frac{\hat{\mu}_i}{\sigma}\right) \\ &\quad + \left(G_i''(Q) - G_i'(Q)^2\right)\frac{1}{2}\sigma_Q^2(Q) \end{aligned}$$

with boundary conditions $G_i'(-\infty) = G_i'(\infty) = 0$. The risk premium satisfies the ODE

$$\begin{aligned} r\Pi(Q) &= -\frac{\phi r \sigma_P^2(Q)}{1-\theta}\left(S - \theta N^Q(Q)\right) + \frac{\bar{\mu}}{r} + \Pi'(Q)\left(\mu_Q(Q) + \sigma_Q(Q)\frac{\bar{\mu}}{\sigma}\right) \\ &\quad + \frac{1}{2}\Pi''(Q)\sigma_Q^2(Q) - \alpha\sigma_P^2(Q)\bar{G}'(Q) \end{aligned}$$

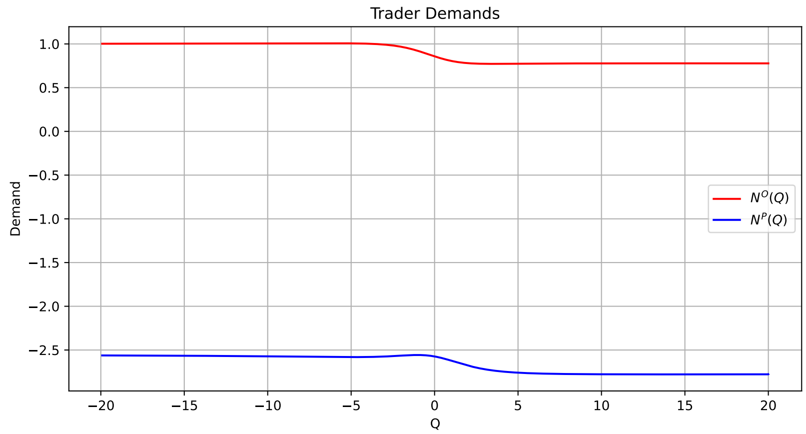
with boundary conditions $\Pi'(-\infty) = \Pi'(\infty) = 0$, and where $\bar{\mu} = m\hat{\mu}_o + (1-m)\hat{\mu}_p$ and $\bar{G}(Q) = mG_O(Q) + (1-m)G_P(Q)$.

Intuitively, optimistic investors demand more of the asset, as they overestimate the div-

[45] The measure $\mathbb{P}$ is defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ satisfying the usual conditions (e.g. Karatzas and Shreve (2014)).

Figure 12: Human traders' demands

The figure plots optimistic and pessimistic traders' equilibrium demands, $N^O(Q)$ and $N^P(Q)$. Parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $m = 0.75$, and $\hat{\mu}_O = 0.1 = -\hat{\mu}_P$.



idend growth. The asset demand of type i is given by

$$N_t^i = \frac{1}{\phi r \sigma_P^2 (Q_t)} \left(\mu_P (Q_t) + \frac{\hat{\mu}_i}{r} + \Pi' (Q_t) \sigma_Q (Q_t) \frac{\hat{\mu}_i}{\sigma} - r\Pi (Q_t) - \frac{\mu}{r} \right) - \frac{\alpha}{\phi r} G_i' (Q_t),$$

which takes a similar form as in the baseline model. The key difference is that type i's subjective drift $\hat{\mu}_i$ enters the demand. The gain $G_i(Q)$ is now the subjective gain under $\mathbb{P}^i$. Type i's actual trading gain is given by

$$V^i (Q_t) = E \left[\int_t^{\infty} e^{-r(s-t)} N_s^i dR_s \right],$$

where the expectation is taken under $\mathbb{P}$, and the Q-learner's profit is given by

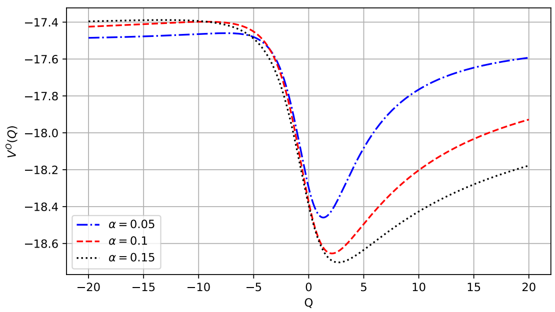
$$U (Q_t) = E \left[\int_t^{\infty} e^{-r(s-t)} N^Q (Q_s) \, dR_s \right].$$

Figure 12 illustrates human traders' equilibrium demands. The optimistic traders buy the asset while the pessimistic traders sell.

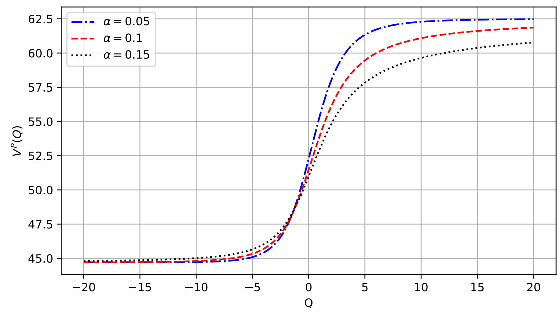
Figure 13 illustrates the equilibrium trading profits for $m > 1/2$. Optimistic traders' profits are negative while pessimistic traders' profits are positive. The Q-learner's profits are positive whenever $Q < 0$ and negative when Q is sufficiently large. Thus, the Q-learner realizes profits by trading against the optimistic traders.

Figure 13: Expected trading profits with disagreement

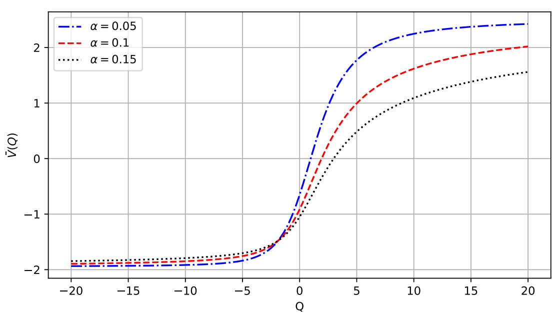
The figure plots the expected trading profits for the Q-learners, $U(Q_t)$, for optimistic and pessimistic traders, $V^O(Q_t)$ and $V^P(Q_t)$, and the average profits of human traders $\bar{V}(Q_t) = mV^O(Q_t) + (1-m)V^P(Q_t)$. Unless specified, parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $m = 0.75$, and $\hat{\mu}_O = 1 = -\hat{\mu}_P$.



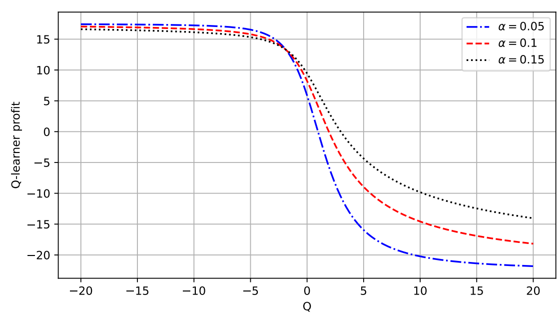
(a) Optimistic traders' profits



(b) Pessimistic traders' profits



(c) Average human traders' profits



(d) Q-learner profits

E.1 Proof of Proposition E.1

The proof proceeds similarly to the proofs of Propositions 2 and A.1. We therefore only provide a sketch. Conjecture that

$$P_t = \frac{1}{r} D_t + \frac{\mu}{r^2} + \Pi(Q_t).$$

Applying Ito's Lemma under $\mathbb{P}^i$ yields[46]

$$dP_t = \left(\mu_P(Q_t) + \frac{\hat{\mu}_i}{r} + \Pi'(Q_t)\,\sigma_Q(Q_t)\,\frac{\hat{\mu}_i}{\sigma} \right) dt + \sigma_P(Q_t)\,dB_t^i$$

[46]Note that the additional drift term $\hat{\mu}_i$ appears in dP_t twice: once via dD_t and once via dQ_t. This reflects the fact that type i has a subjective belief about both the evolution of prices and the evolution of Q-values.

We denote with $\mu_P(Q_t)$ the drift of P_t under $\mathbb{P}$ (i.e. Equation (17)) to keep the effect of disagreement separate. Using Equation (15) and Ito's Lemma under $\mathbb{P}^i$ similarly yields

$$dQ_t = \left(\mu_Q(Q_t) + \alpha\Pi'(Q_t)\sigma_Q(Q_t)\frac{\hat{\mu}_i}{\sigma}\right)dt + \sigma_Q(Q_t)\,dB_t^i.$$

A similar derivation as in Section 3 then implies that the drift of Q_t under $\mathbb{P}^i$ is

$$\mu_Q(Q_t)+\alpha\Pi'(Q_t)\sigma_Q(Q_t)\frac{\hat{\mu}_i}{\sigma} = \alpha\left(\Pi'(Q_t)\left(\mu_Q(Q_t)+\sigma_Q(Q_t)\frac{\hat{\mu}_i}{\sigma}\right)+\frac{1}{2}\Pi''(Q_t)\sigma_Q^2(Q_t)-r\Pi(Q_t)-\frac{1}{2}Q_t\right),$$

the drift of P_t under $\mathbb{P}^i$ is

$$\mu_P(Q_t) = \frac{\mu}{r}+\frac{\alpha\Pi'(Q_t)}{1-\alpha\Pi'(Q_t)}\left(\frac{1}{2}\Pi''(Q_t)\sigma_Q^2(Q_t)-r\Pi(Q_t)-\frac{1}{2}Q_t\right)+\frac{1}{2}\Pi''(Q_t)\sigma_Q^2(Q_t),$$

and the volatility terms are

$$\sigma_Q(Q_t) = \alpha\sigma_P(Q_t) \text{ and } \sigma_P(Q_t) = \frac{1}{1-\alpha\Pi'(Q_t)}\frac{\sigma}{r}.$$

Finally, type i's wealth process under $\mathbb{P}^i$ follows

$$dW_t = \left(rW_t - C_t + N_t\left(\mu_P(Q_t)+\frac{\hat{\mu}_i}{r}+\Pi'(Q_t)\sigma_Q(Q_t)\frac{\hat{\mu}_i}{\sigma}-r\Pi(Q_t)-\frac{\mu}{r}\right)\right)dt + N_t\sigma_P(Q_t)\,dB_t^i.$$

Similar to the proof of Proposition A.1, we conjecture that type i's value function is given by

$$V^i(W,Q) = -\frac{1}{\phi r}\exp\left(-\phi rW - G_i(Q)-\frac{\delta-r}{r}\right).$$

The HJB equation

$$\begin{aligned}
\delta V^i = \; & \max_{C,N} -\frac{1}{\phi}e^{-\phi C} + V_W^i\left(rW - C + N\left(\mu_P(Q)+\frac{\hat{\mu}_i}{r}+\Pi'(Q)\sigma_Q(Q)\frac{\hat{\mu}_i}{\sigma}-r\Pi(Q)-\frac{\mu}{r}\right)\right)\\
& +\frac{1}{2}V_{WW}^i\sigma_P^2(Q)N^2\\
& +V_Q^i\left(\mu_Q(Q)+\alpha\Pi'(Q)\sigma_Q(Q)\frac{\hat{\mu}_i}{\sigma}\right)+V_{QQ}\frac{1}{2}\sigma_Q^2(Q)\\
& +V_{WQ}^iN\sigma_P(Q)\sigma_Q(Q).
\end{aligned}$$

then yields the asset demand

$$N_t^i = \frac{1}{\phi r\sigma_P^2(Q_t)}\left(\mu_P(Q_t)+\frac{\hat{\mu}_i}{r}+\Pi'(Q_t)\sigma_Q(Q_t)\frac{\hat{\mu}_i}{\sigma}-r\Pi(Q_t)-\frac{\mu}{r}\right)-\frac{\alpha}{\phi r}G_i'(Q_t).$$

Using the market clearing condition

$$\theta N^Q(Q_t)+(1-\theta)\left(mN_t^O+(1-m)N_t^P\right) = S$$

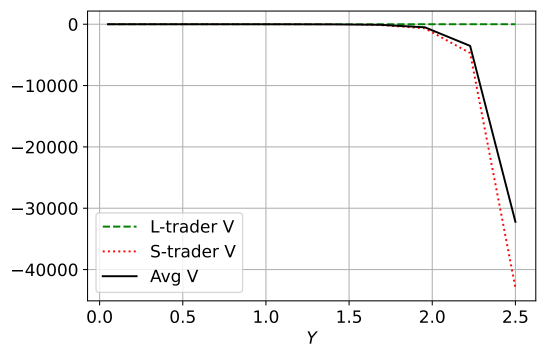
and a similar derivation as in the proof of Proposition 2 then yields the system of ODEs in Equations (47) and (48).

F Supplementary figures

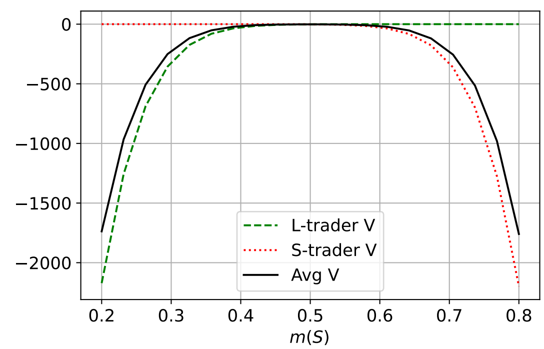
F.1 Endowment risk: Expected value for rational investors

Figure 14: Expected Value for rational investors

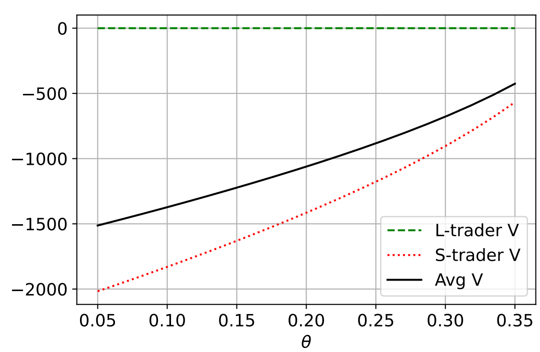
The figure plots the expected utility for L and S investors, V_L (dashed) and V_S (dotted), and the average gain $V = mV_S + (1-m)V_L$ (solid), versus various model parameters. Unless specified, parameters are set to: $\phi = 10$, $\sigma = 0.075$, $r = 0.05$, $\beta = 1$, $\mu = 0.05$, $\theta = 0.30$, $\alpha = 0.10$, $S = 0$, $m = 0.75$ and $Y = 2$.



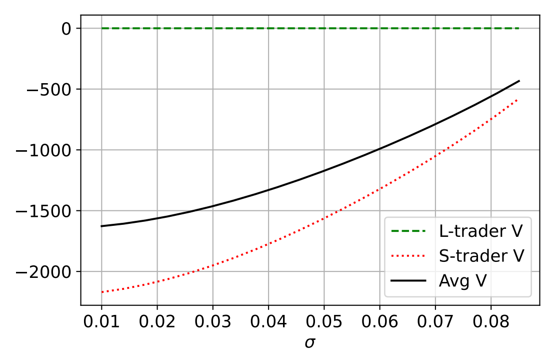
(a) Investor value versus Y



(b) Investor value versus m



(c) Investor value versus θ



(d) Investor value versus σ